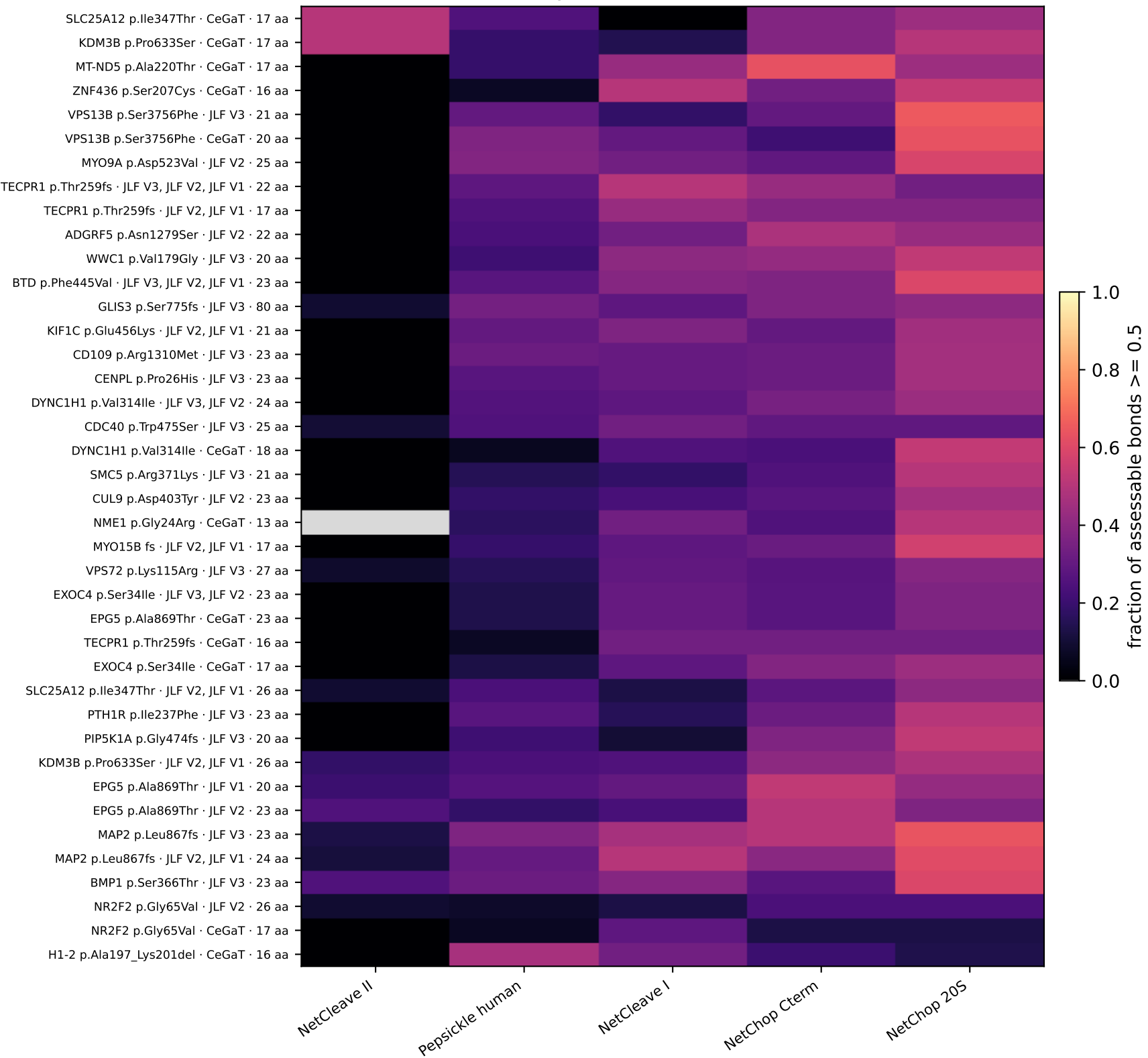
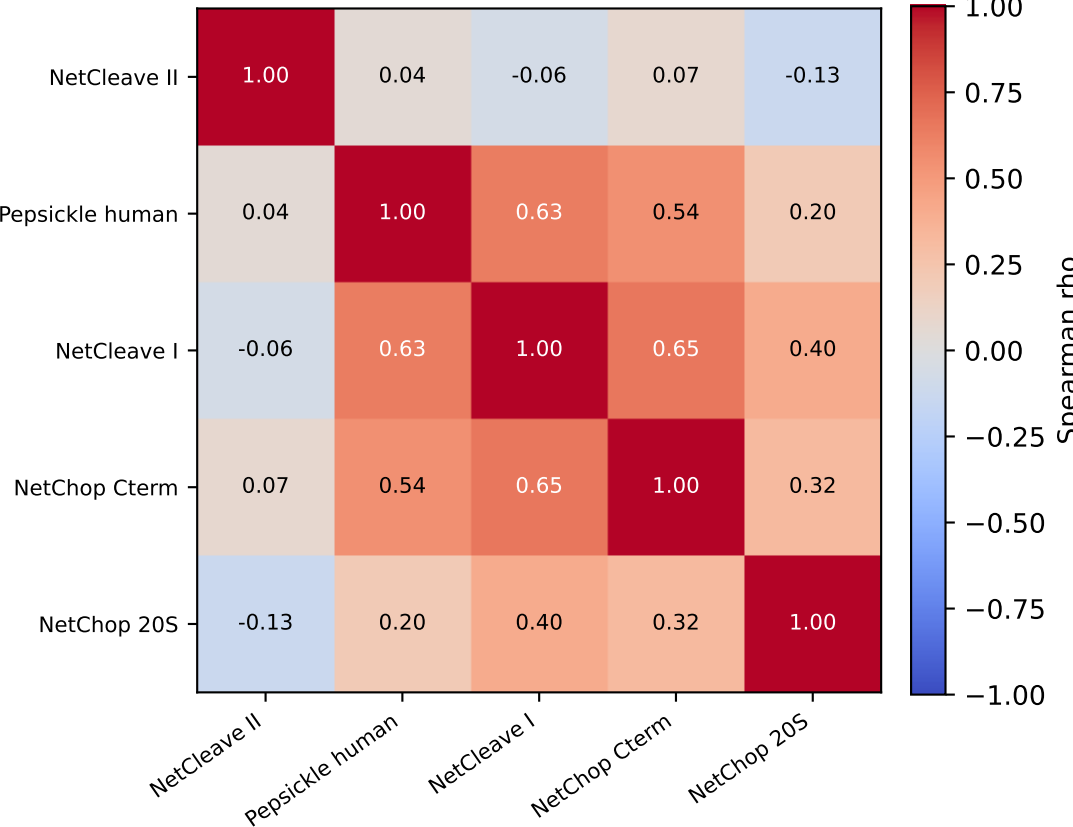


Agreement map for the vaccine SLP cleavage atlas

SLPs clustered by within-model candidate-site fraction



Predictor agreement at shared peptide bonds



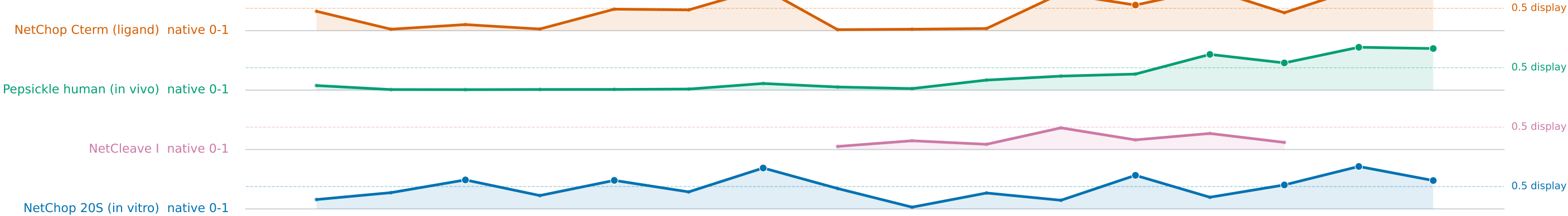
Clustering organizes the atlas; it does not create an ensemble probability. Sequence pages follow the left-panel SLP order. The figure shows human-only Pepsickle, both biologically distinct NetChop modes, NetCleave-I, and NetCleave-II; the near-redundant all-mammal Pepsickle track remains in the tables. The left panel uses a common display threshold on native 0-1 outputs. The right panel correlates native scores only where both models assess the same bond.

SLC25A12 p.Ile347Thr | CeGaT | 17 aa

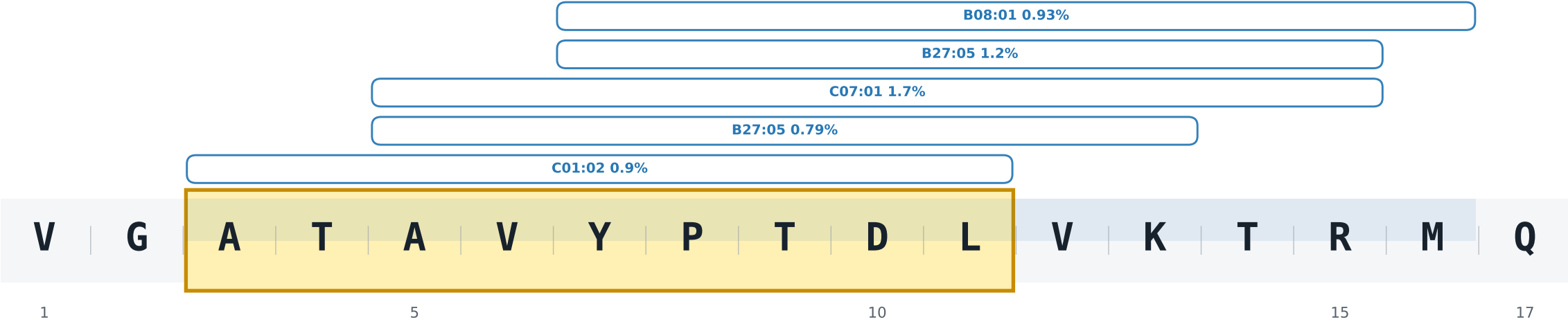
disclosed minimal epitope ATAVYPTDL at 3-11

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



MHC-I: 5 shown / 18 eligible (top-10 cap)

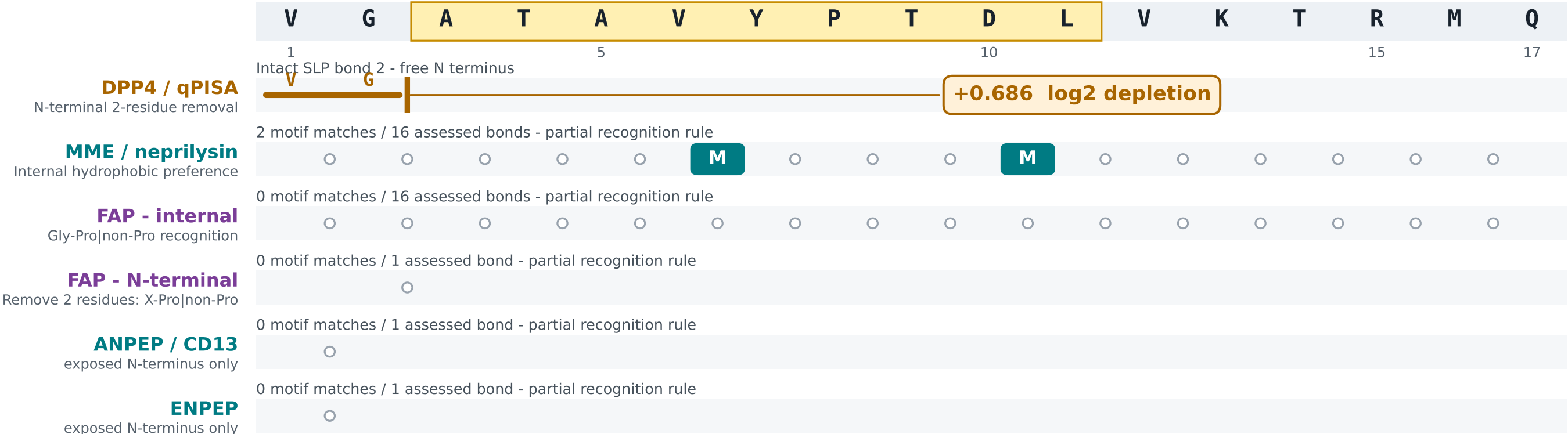


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

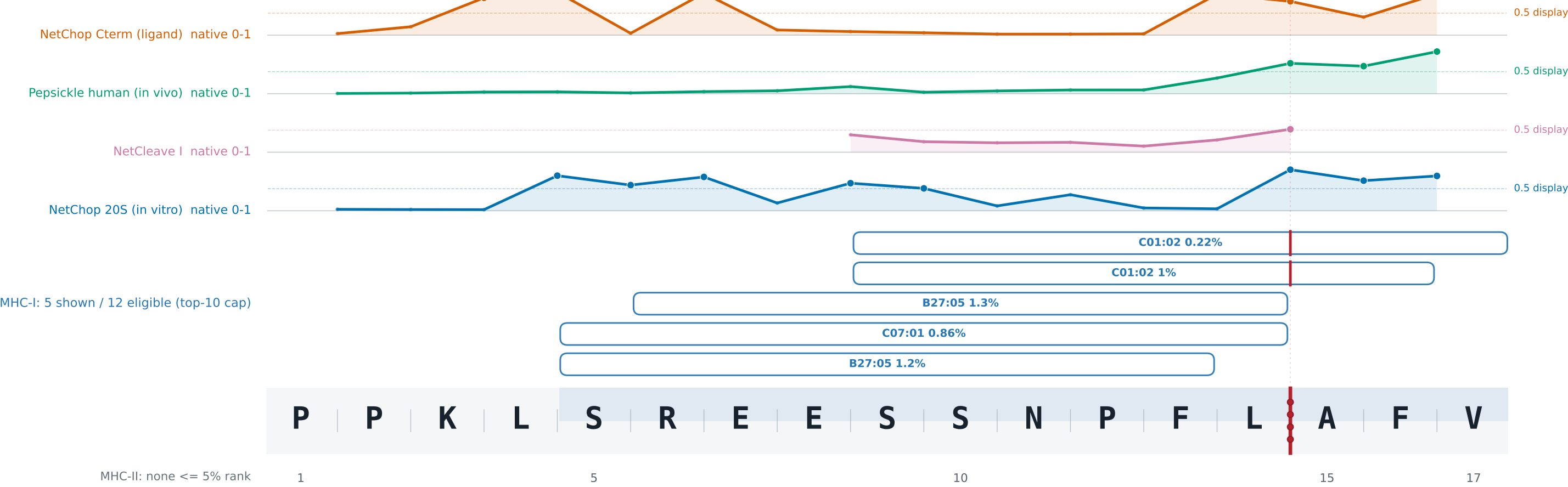
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

KDM3B p.Pro633Ser | CeGaT | 17 aa

minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

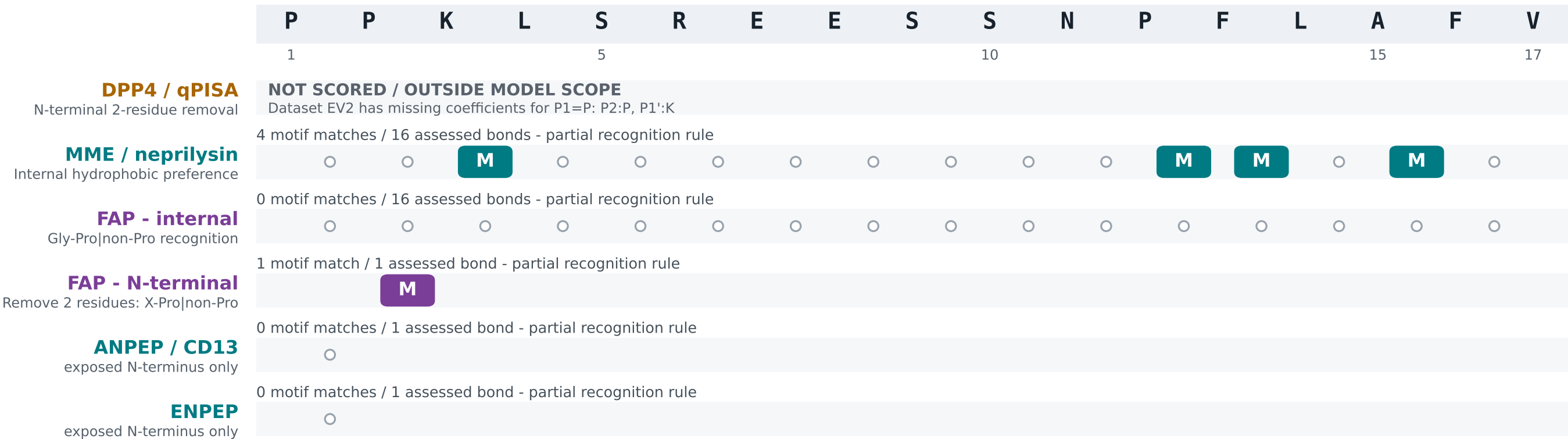


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

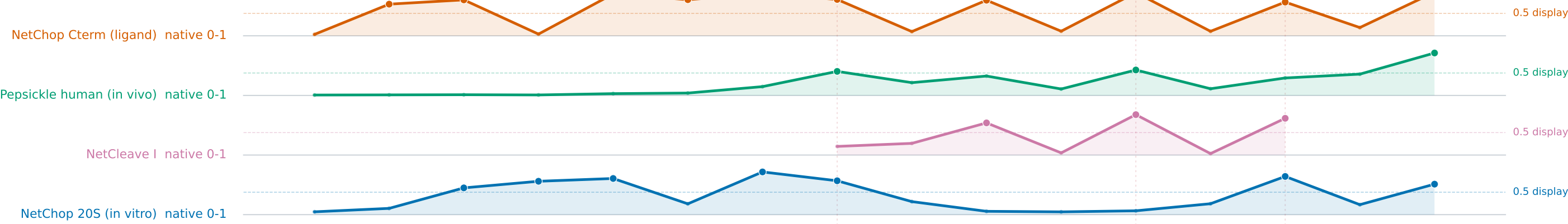
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

MT-ND5 p.Ala220Thr | CeGaT | 17 aa

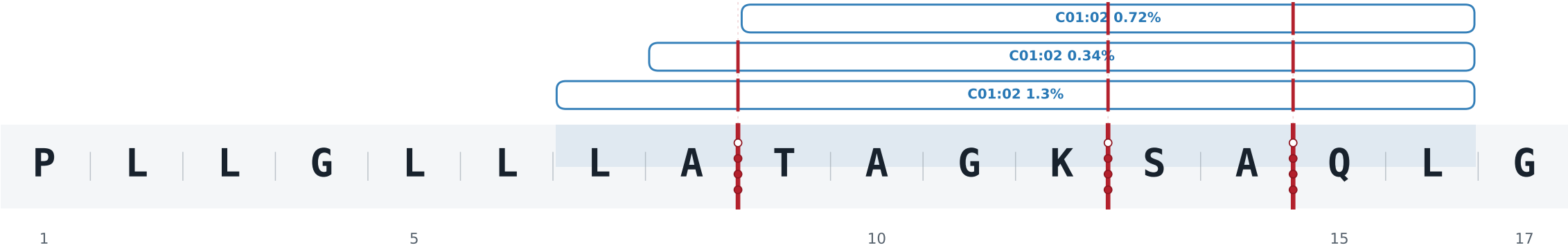
minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

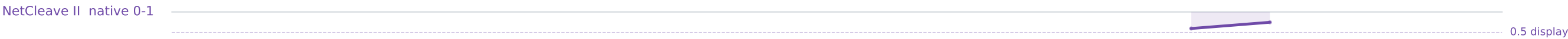


MHC-I: 3 shown / 5 eligible (top-10 cap)

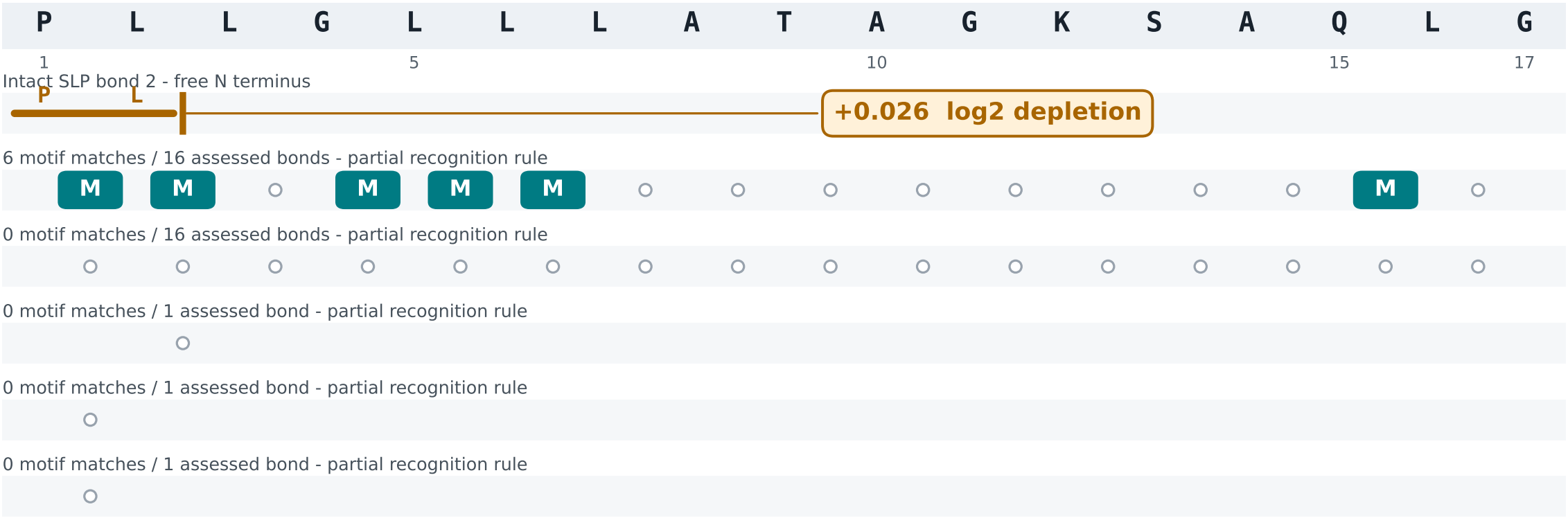


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

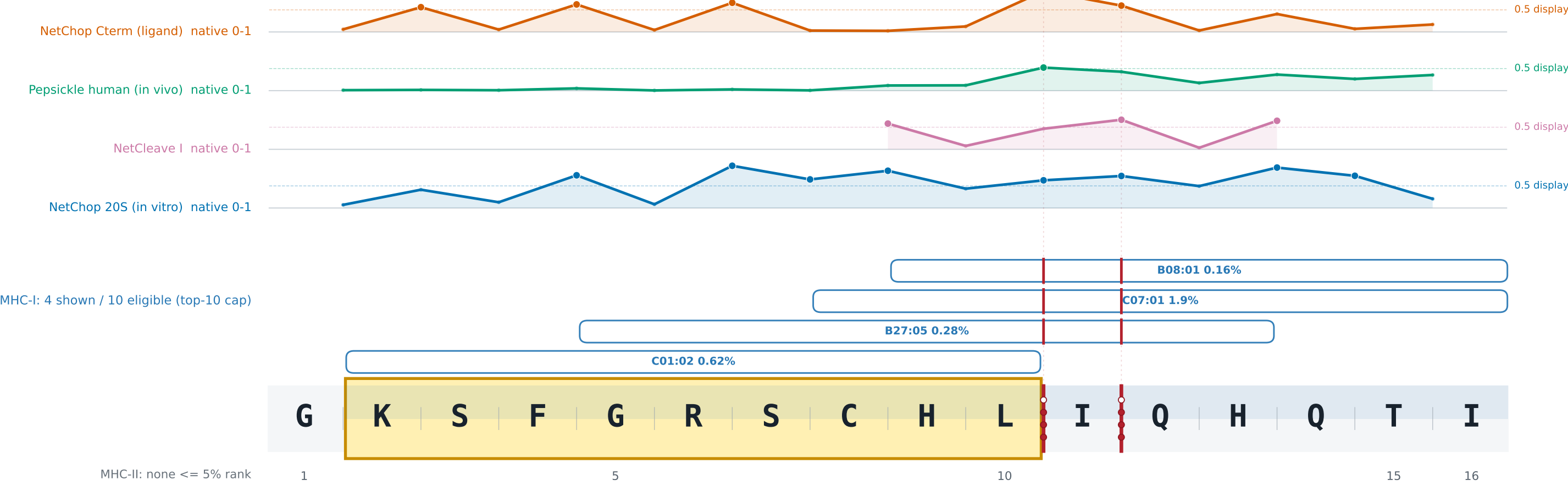
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

ZNF436 p.Ser207Cys | CeGaT | 16 aa

disclosed minimal epitope KSFGRSCHL at 2-10

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

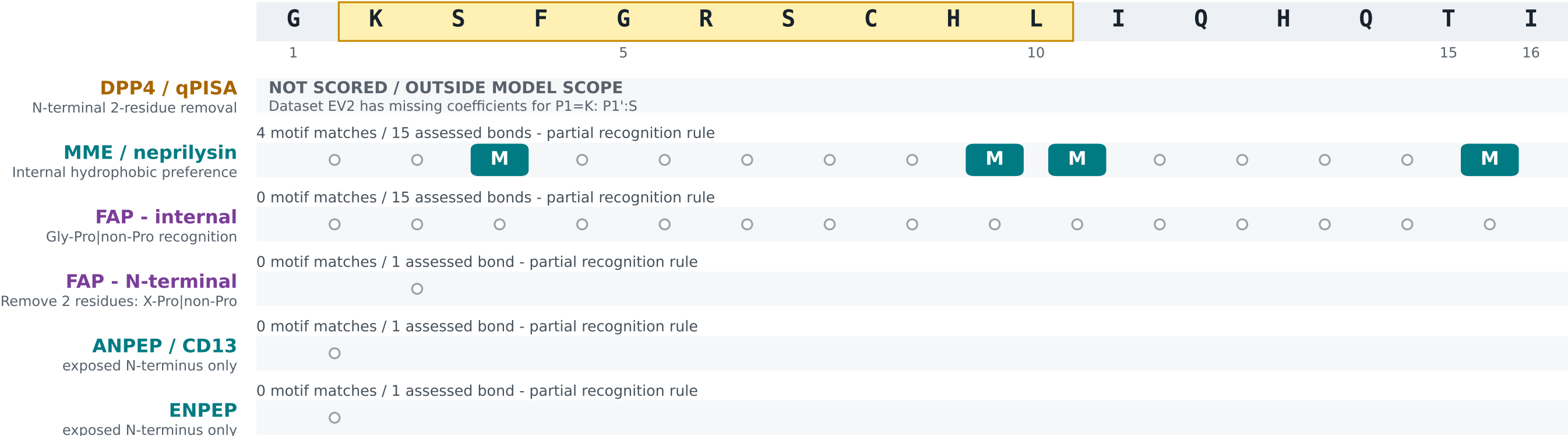


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

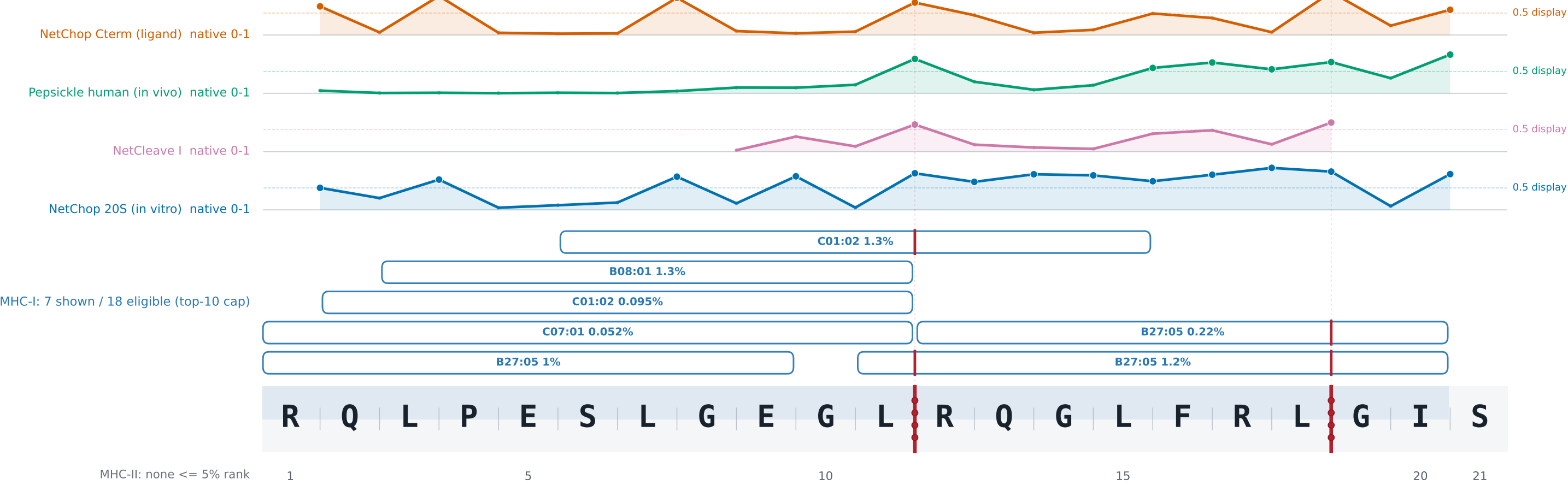
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

VPS13B p.Ser3756Phe| JLF V3 | 21 aa

minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

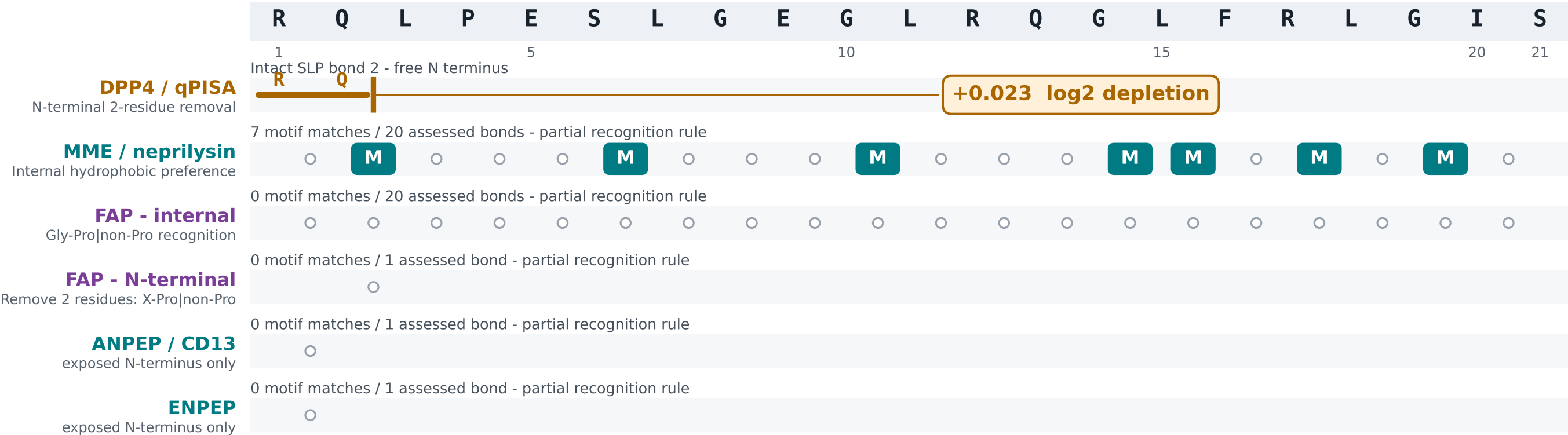


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

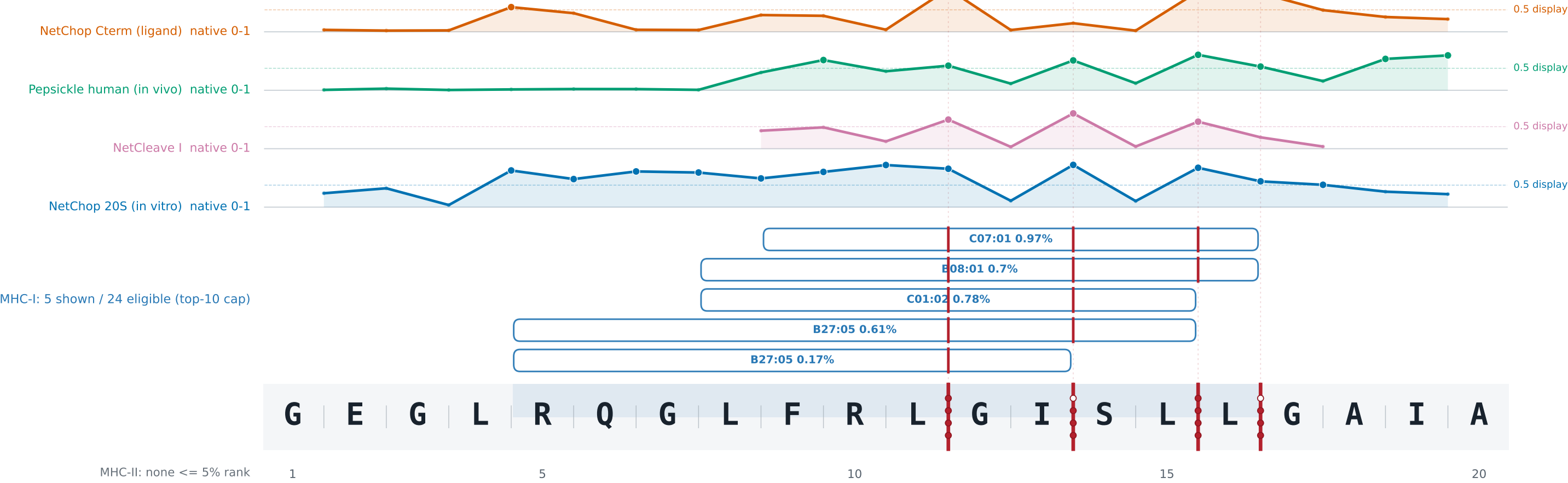
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

VPS13B p.Ser3756Phe| CeGaT | 20 aa

minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

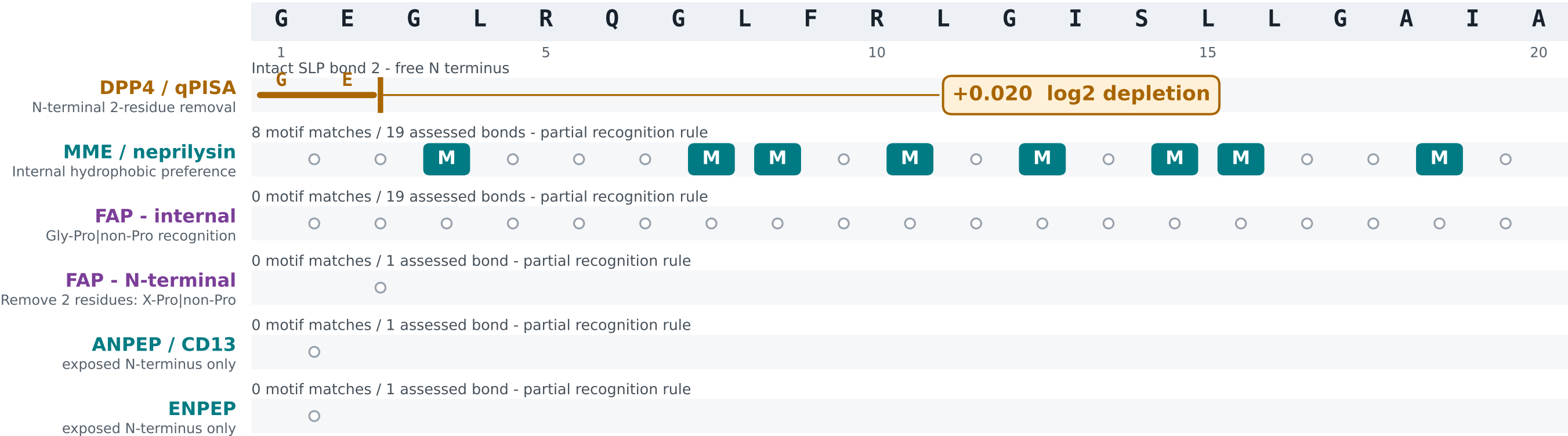


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

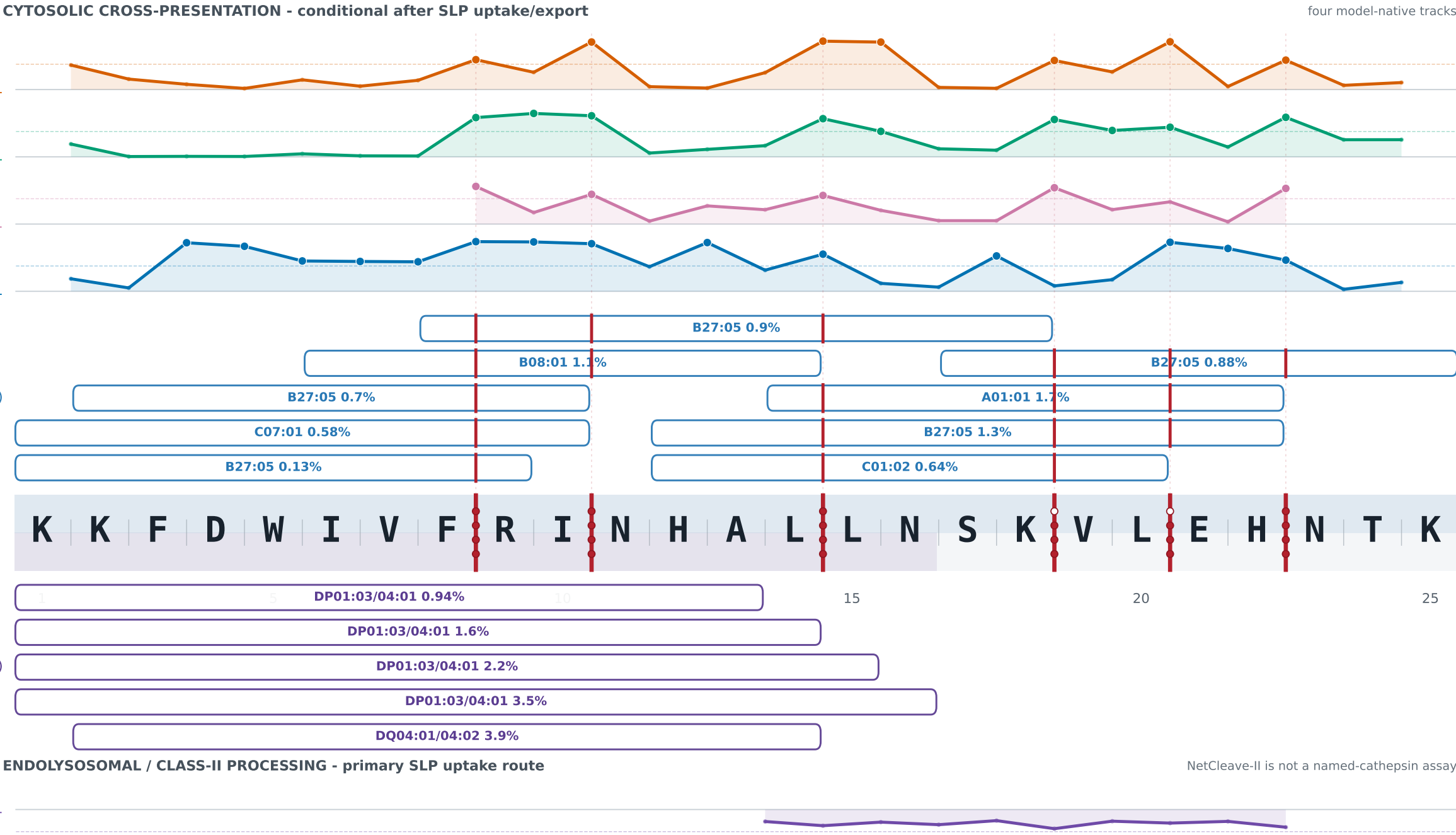
SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

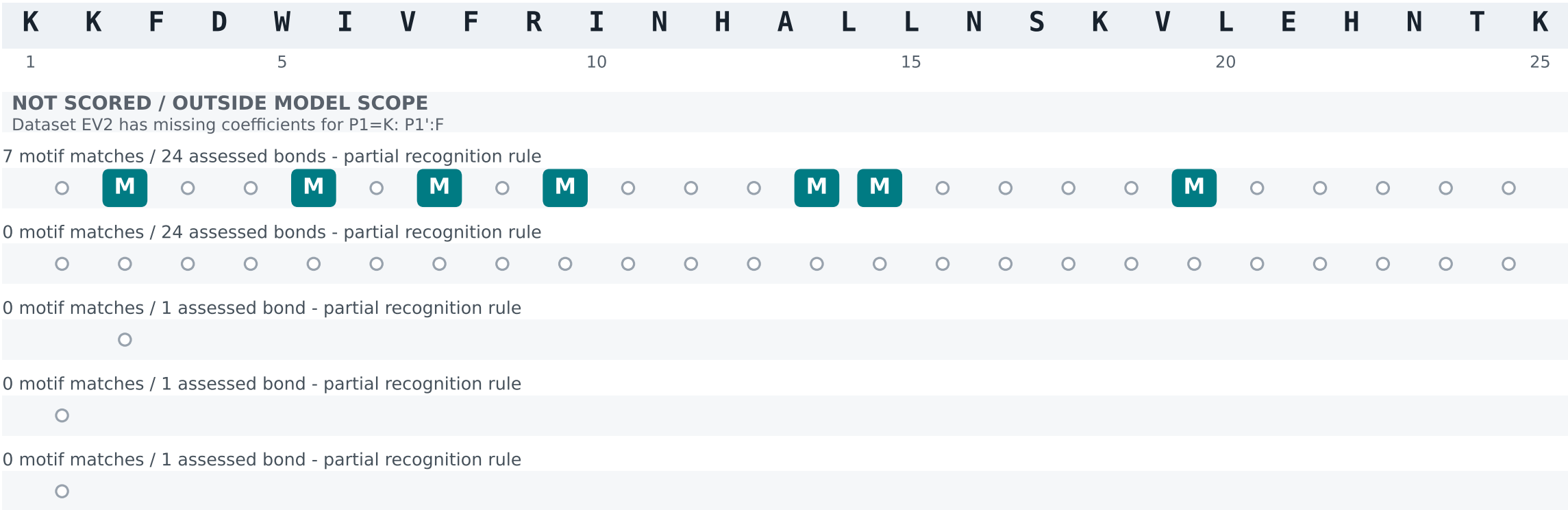
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

MYO9A p.Asp523Val | JLF V2 | 25 aa

minimal epitope/window not disclosed for this SLI



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

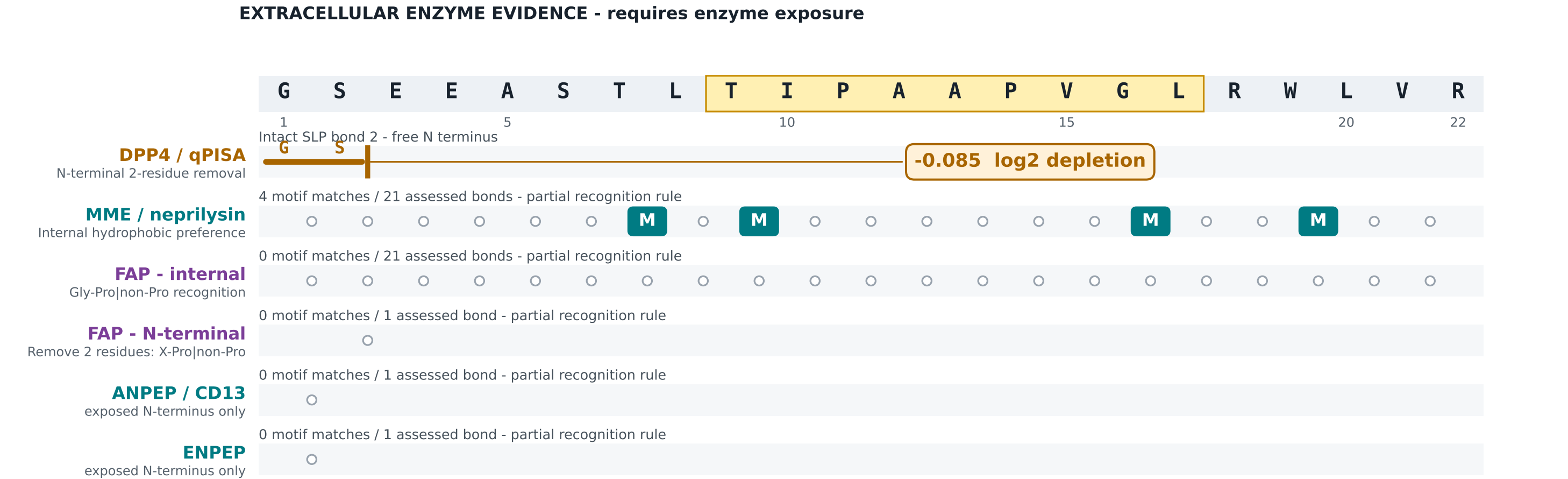
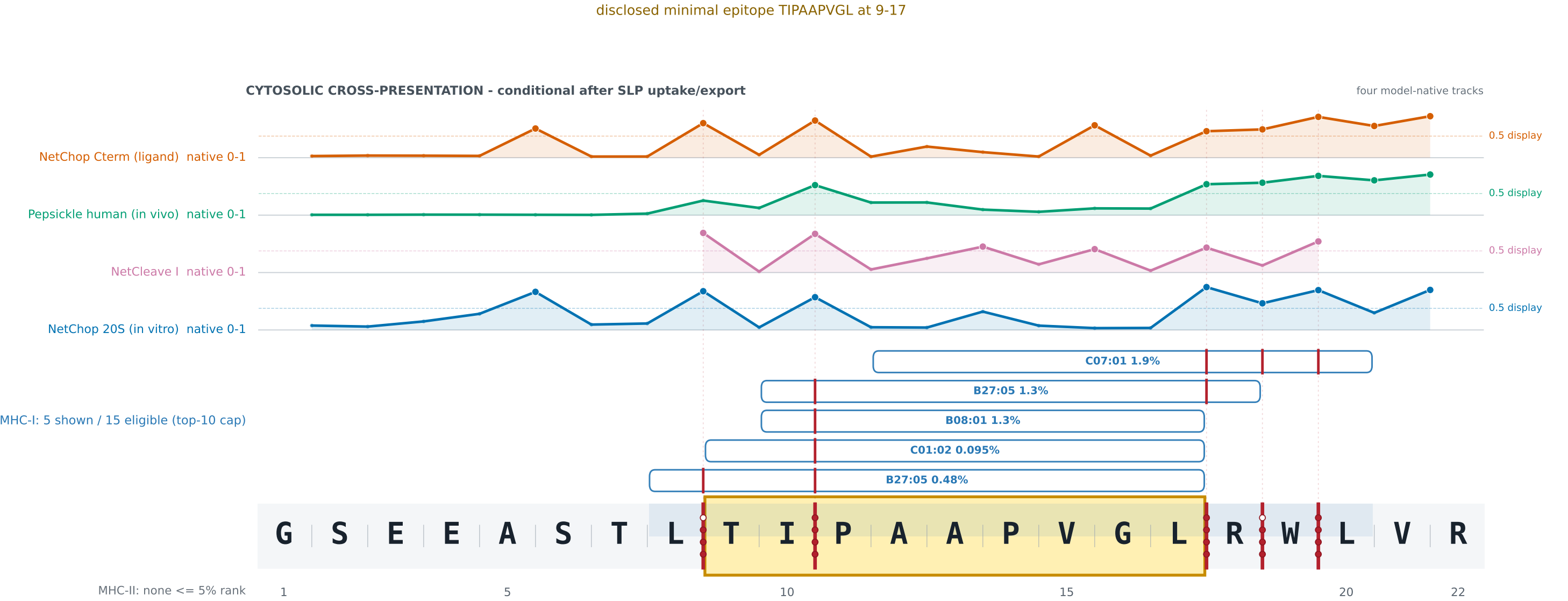
DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed

RED All four cytosolic tracks assessed the bond and ≥ 3 reached 0.5. Four filled/open beads show model concurrence, never probability

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

TECPR1 p.Thr259fs | JLF V3, JLF V2, JLF V1 | 22 aa



M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

TECPR1 p.Thr259fs | JLF V2, JLF V1 | 17 aa

minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

NetChop Cterm (ligand) native 0-1

Pepsickle human (in vivo) native 0-1

NetCleave I native 0-1

NetChop 20S (in vitro) native 0-1

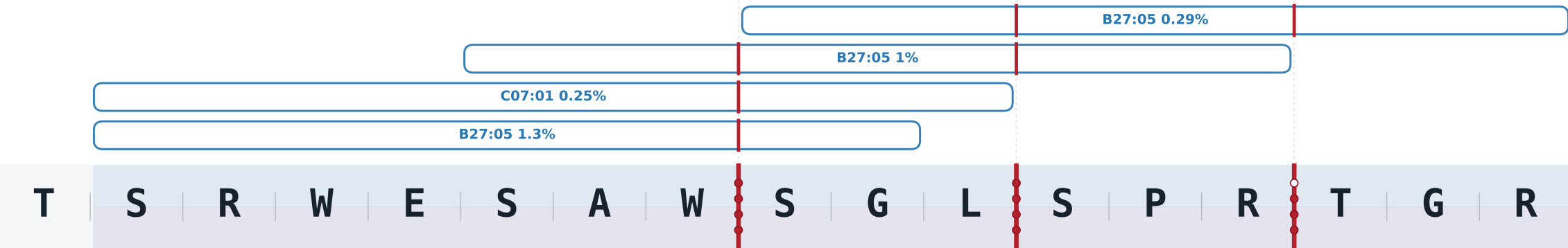
0.5 display

0.5 display

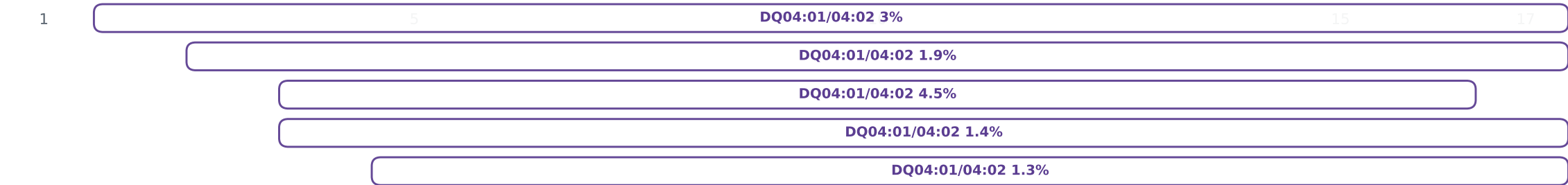
0.5 display

0.5 display

MHC-I: 4 shown / 5 eligible (top-10 cap)



MHC-II: 5 shown / 5 eligible (top-10 cap)



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay

NetCleave II native 0-1

0.5 display

EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure

T	S	R	W	E	S	A	W	S	G	L	S	P	R	T	G	R
1				5					10					15		17

DPP4 / qPISA
N-terminal 2-residue removal

NOT SCORED / OUTSIDE MODEL SCOPE
Dataset EV2 has missing coefficients for P1=S: P1':R

1 motif match / 16 assessed bonds - partial recognition rule

○ ○ ○ ○ ○ ○ ○ ○ ○ ○ M ○ ○ ○ ○ ○ ○ ○

MME / neprilysin
Internal hydrophobic preference

0 motif matches / 16 assessed bonds - partial recognition rule

○ ○ ○ ○ ○ ○ ○ ○ ○ ○ ○ ○ ○ ○ ○ ○ ○

FAP - internal
Gly-Pro|non-Pro recognition

0 motif matches / 1 assessed bond - partial recognition rule

○

FAP - N-terminal
Remove 2 residues: X-Pro|non-Pro

0 motif matches / 1 assessed bond - partial recognition rule

○

ANPEP / CD13
exposed N-terminus only

0 motif matches / 1 assessed bond - partial recognition rule

○

ENPEP
exposed N-terminus only

M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

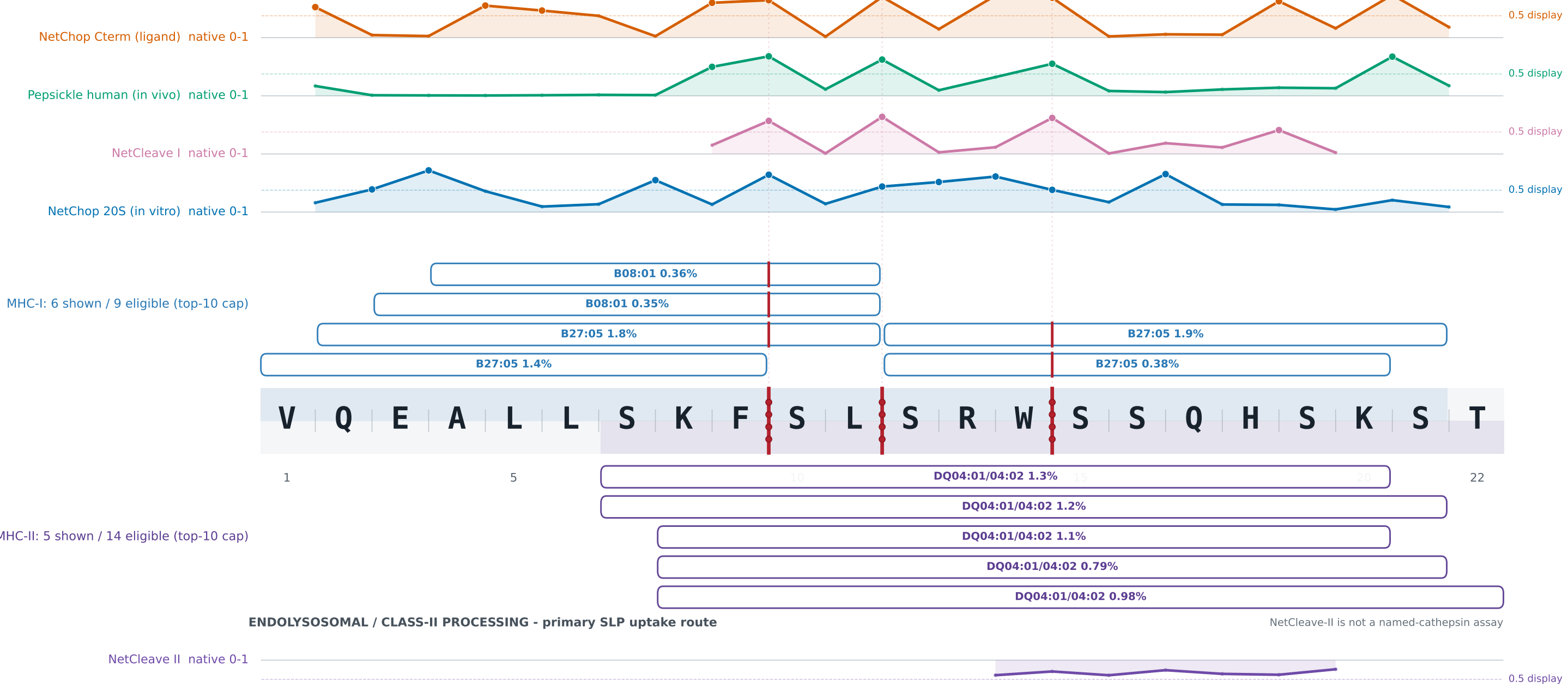
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

ADGRF5 p.Asn1279Ser | JLF V2 | 22 aa

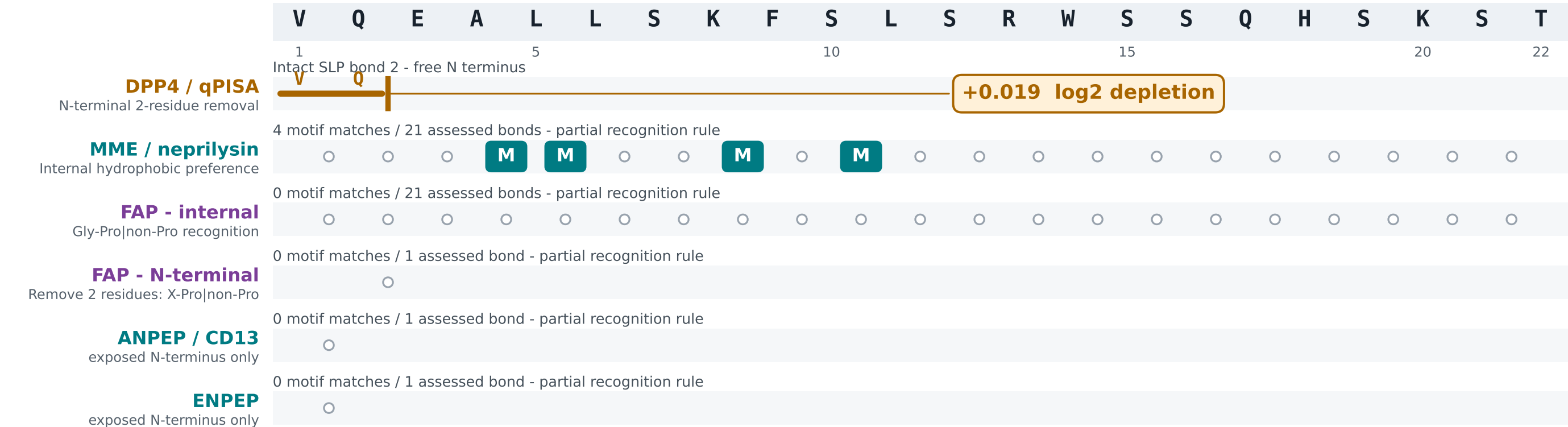
minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.
DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

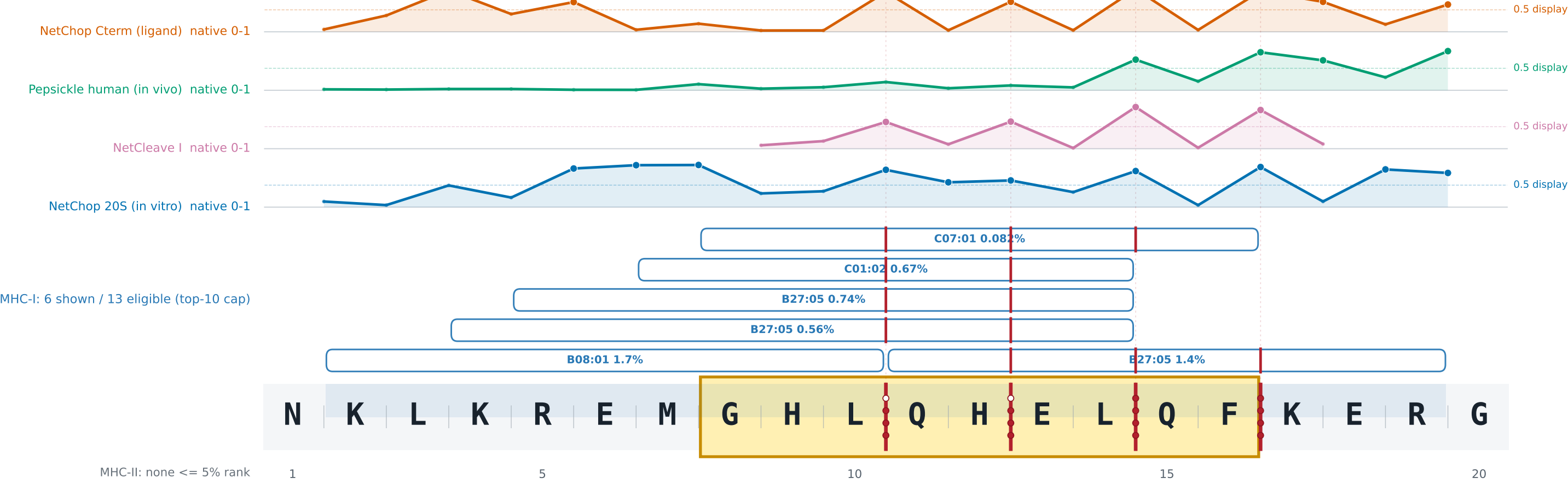
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

WWC1 p.Val179Gly | JLF V3 | 20 aa

disclosed minimal epitope GHLQHELQF at 8-16

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

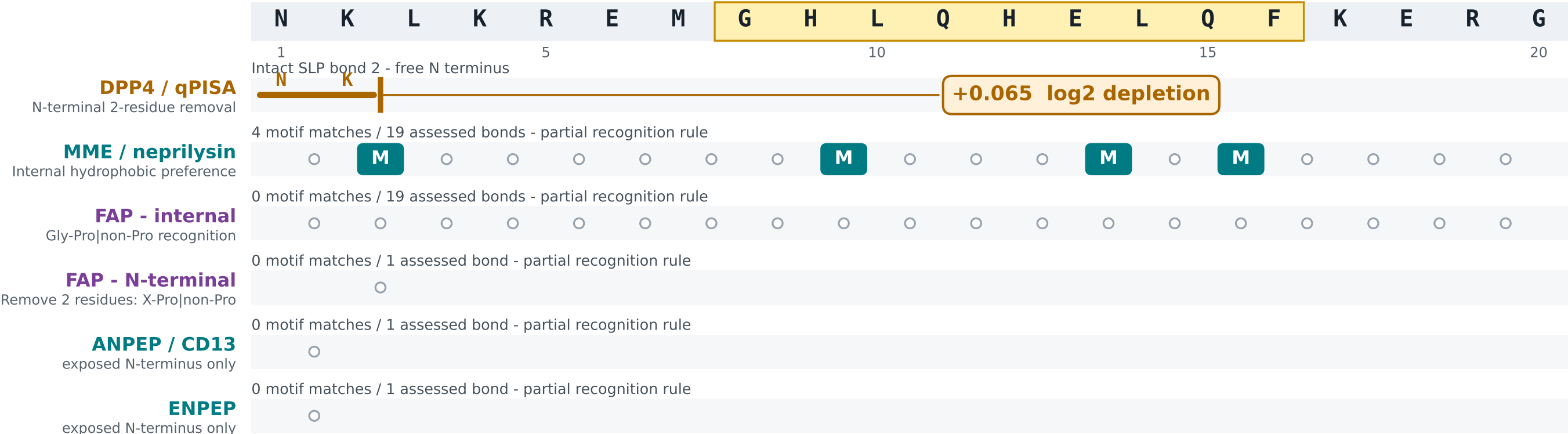


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

BTD p.Phe445Val | JLF V3, JLF V2, JLF V1 | 23 aa

disclosed minimal epitope GVVDGLHTV at 12-20

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

NetChop Cterm (ligand) native 0-1

Pepsickle human (in vivo) native 0-1

NetCleave I native 0-1

NetChop 20S (in vitro) native 0-1

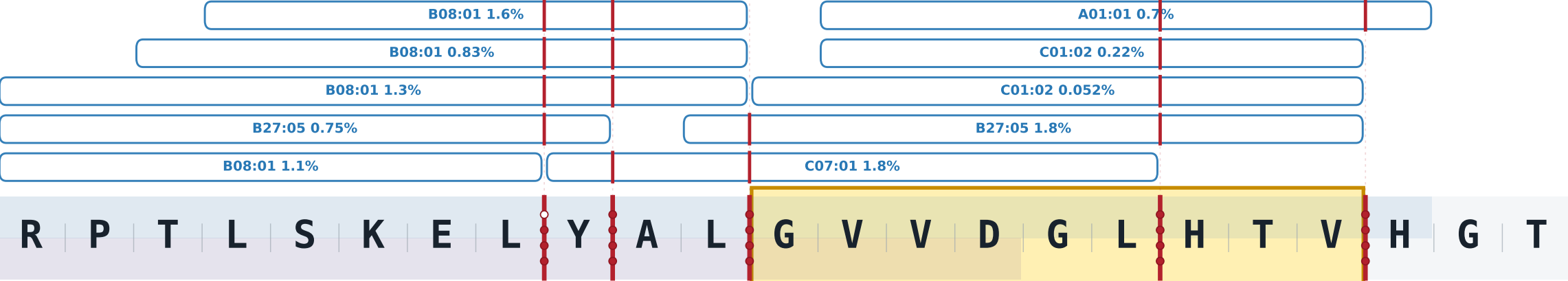
0.5 display

0.5 display

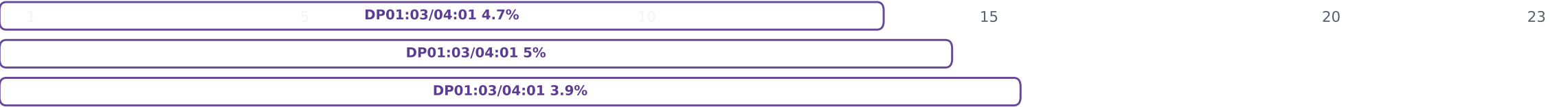
0.5 display

0.5 display

MHC-I: 10 shown / 20 eligible (top-10 cap)



MHC-II: 3 shown / 3 eligible (top-10 cap)



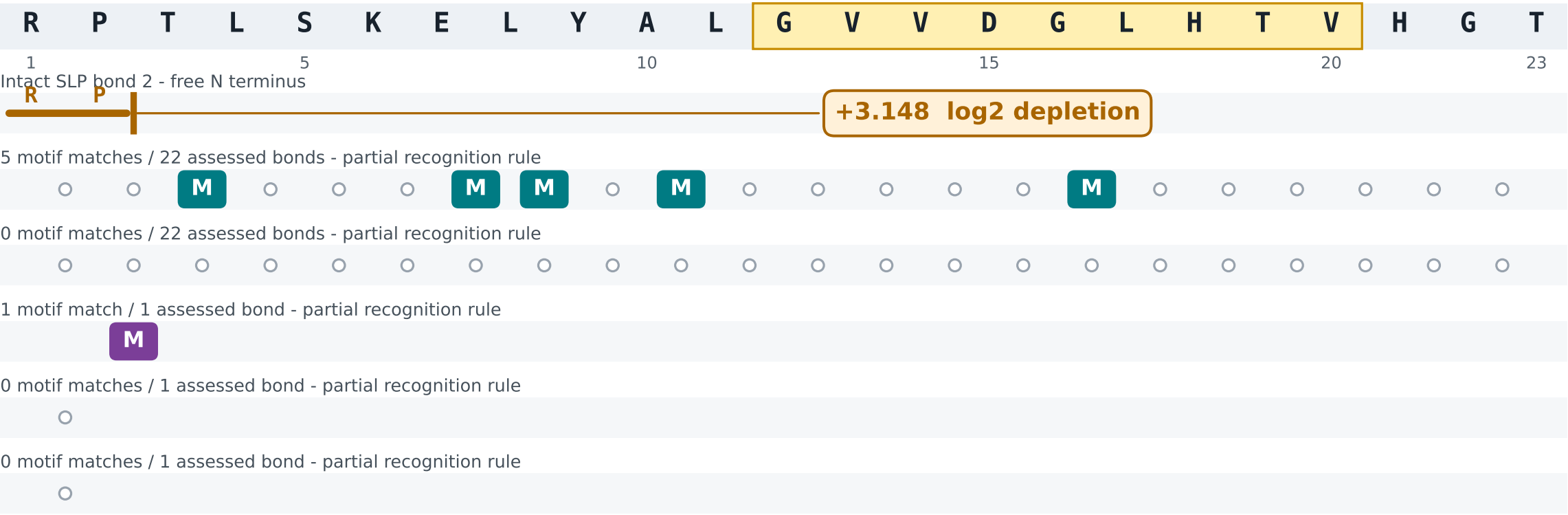
ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay

NetCleave II native 0-1

0.5 display

EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

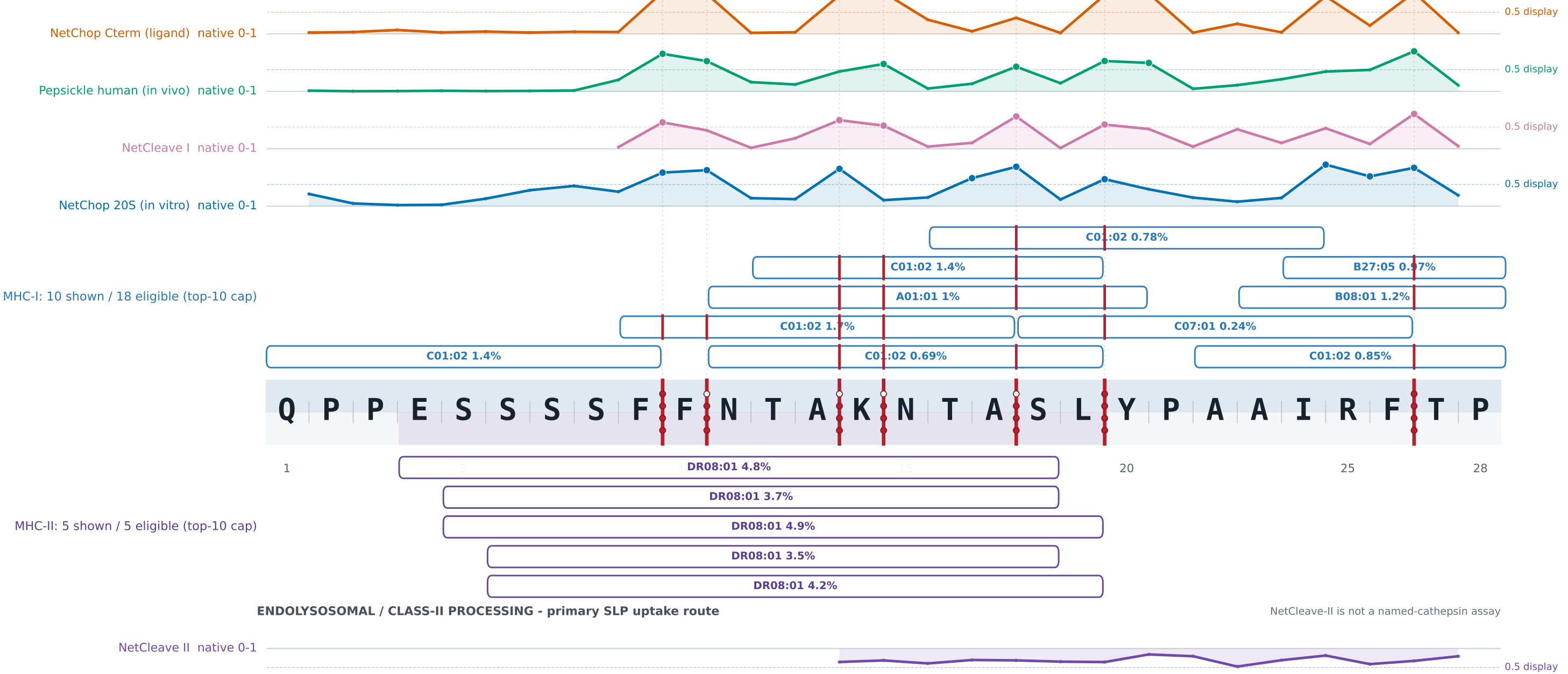
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

GLIS3 p.Ser775fs | JLF V3 | 80 aa - segment 1/3, bonds 1-27

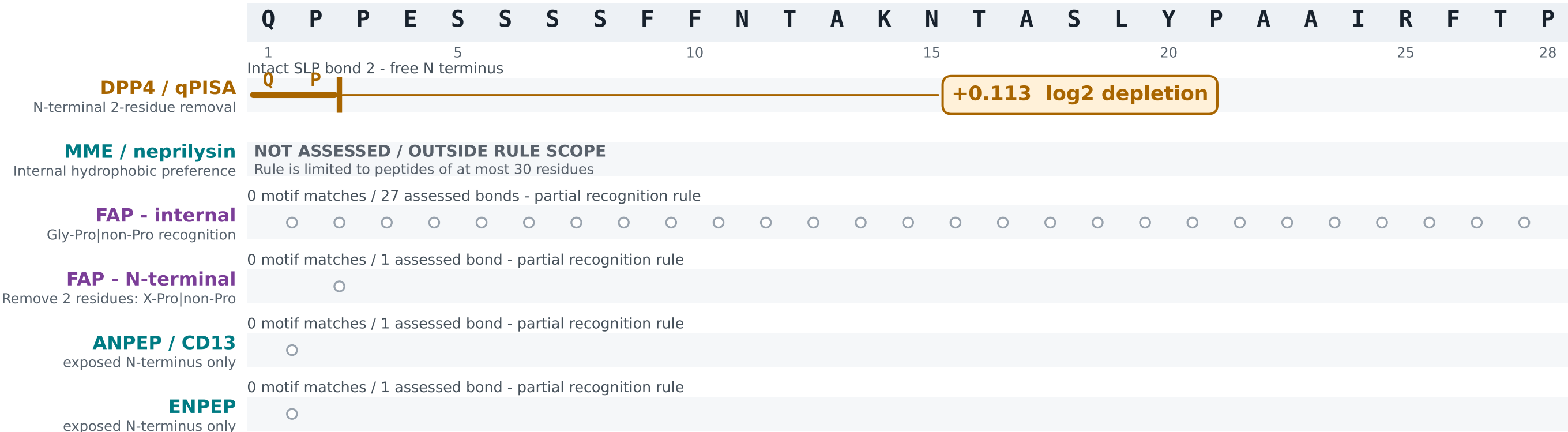
minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

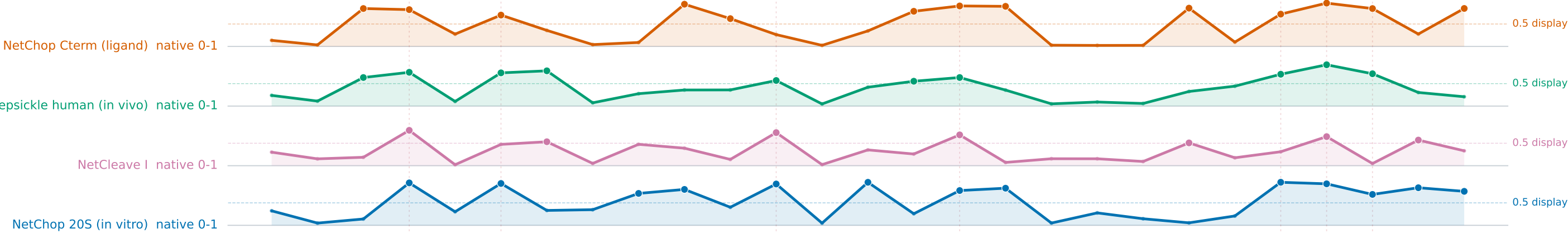
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

GLIS3 p.Ser775fs | JLF V3 | 80 aa - segment 2/3, bonds 28-54

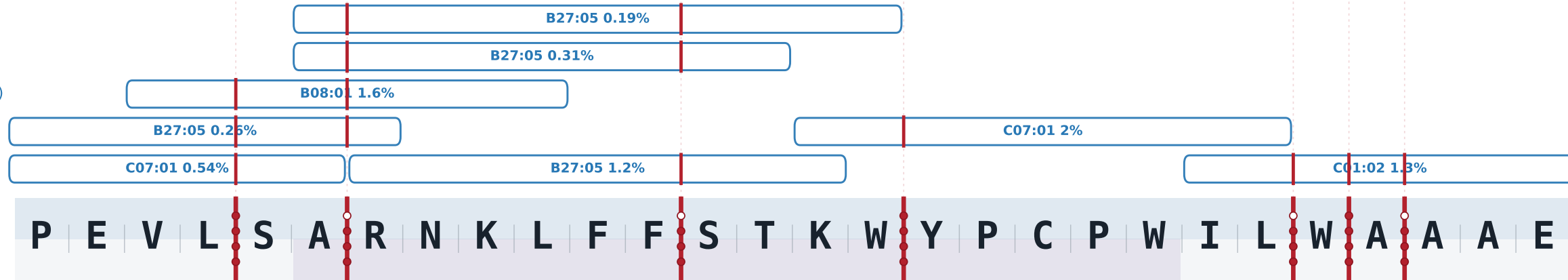
minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

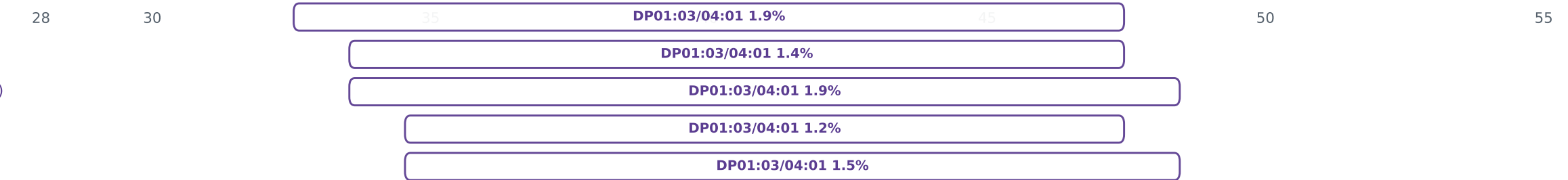
four model-native tracks



MHC-I: 8 shown / 14 eligible (top-10 cap)



MHC-II: 5 shown / 19 eligible (top-10 cap)



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



DPP4 / qPISA
N-terminal 2-residue removal

SITE OUTSIDE THIS SEGMENT | full-SLP score +0.113 log2 depletion
Intact SLP bond 2 is outside this segment

MME / neprilysin
Internal hydrophobic preference

NOT ASSESSED / OUTSIDE RULE SCOPE
Rule is limited to peptides of at most 30 residues

FAP - internal
Gly-Pro|non-Pro recognition

0 motif matches / 27 assessed bonds - partial recognition rule

○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

FAP - N-terminal
Remove 2 residues: X-Pro|non-Pro

SITE OUTSIDE THIS SEGMENT
Intact N-terminal bond 2 is outside this segment

ANPEP / CD13
exposed N-terminus only

SITE OUTSIDE THIS SEGMENT
Intact N-terminal bond 1 is outside this segment

ENPEP
exposed N-terminus only

SITE OUTSIDE THIS SEGMENT
Intact N-terminal bond 1 is outside this segment

M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.
DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

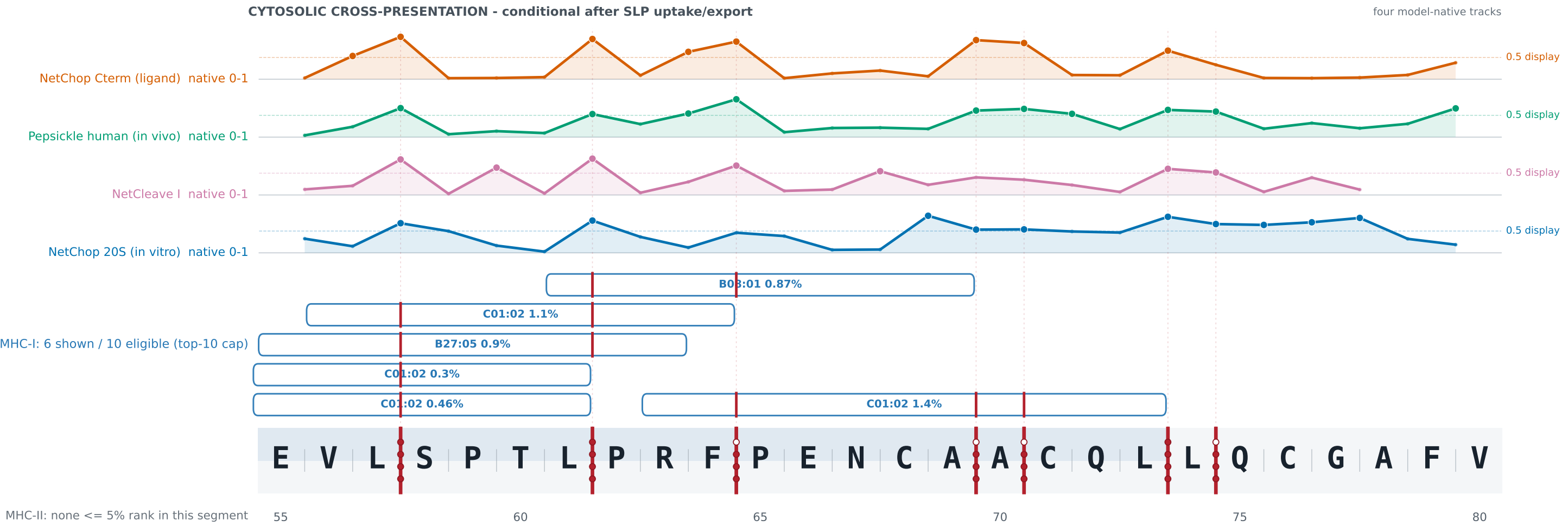
SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

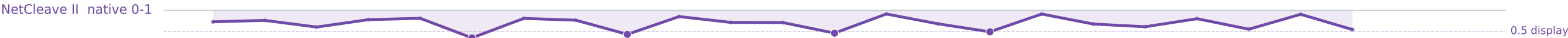
GLIS3 p.Ser775fs | JLF V3 | 80 aa - segment 3/3, bonds 55-79

minimal epitope/window not disclosed for this SLP



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



DPP4 / qPISA N-terminal 2-residue removal	SITE OUTSIDE THIS SEGMENT full-SLP score +0.113 log2 depletion Intact SLP bond 2 is outside this segment
MME / neprilysin Internal hydrophobic preference	NOT ASSESSED / OUTSIDE RULE SCOPE Rule is limited to peptides of at most 30 residues
FAP - internal Gly-Pro non-Pro recognition	0 motif matches / 25 assessed bonds - partial recognition rule
FAP - N-terminal Remove 2 residues: X-Pro non-Pro	SITE OUTSIDE THIS SEGMENT Intact N-terminal bond 2 is outside this segment
ANPEP / CD13 exposed N-terminus only	SITE OUTSIDE THIS SEGMENT Intact N-terminal bond 1 is outside this segment
ENPEP exposed N-terminus only	SITE OUTSIDE THIS SEGMENT Intact N-terminal bond 1 is outside this segment

M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

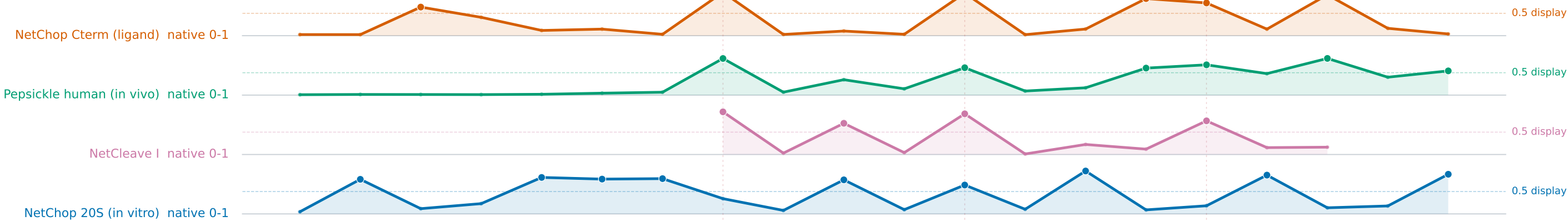
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

KIF1C p.Glu456Lys | JLF V2, JLF V1 | 21 aa

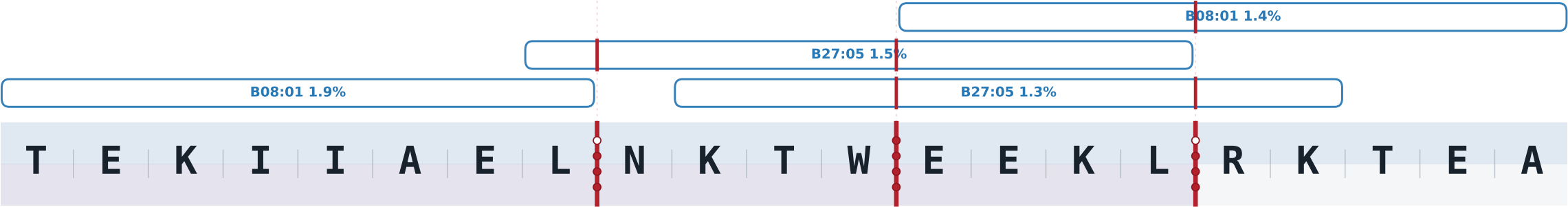
minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



MHC-I: 4 shown / 4 eligible (top-10 cap)



MHC-II: 5 shown / 8 eligible (top-10 cap)



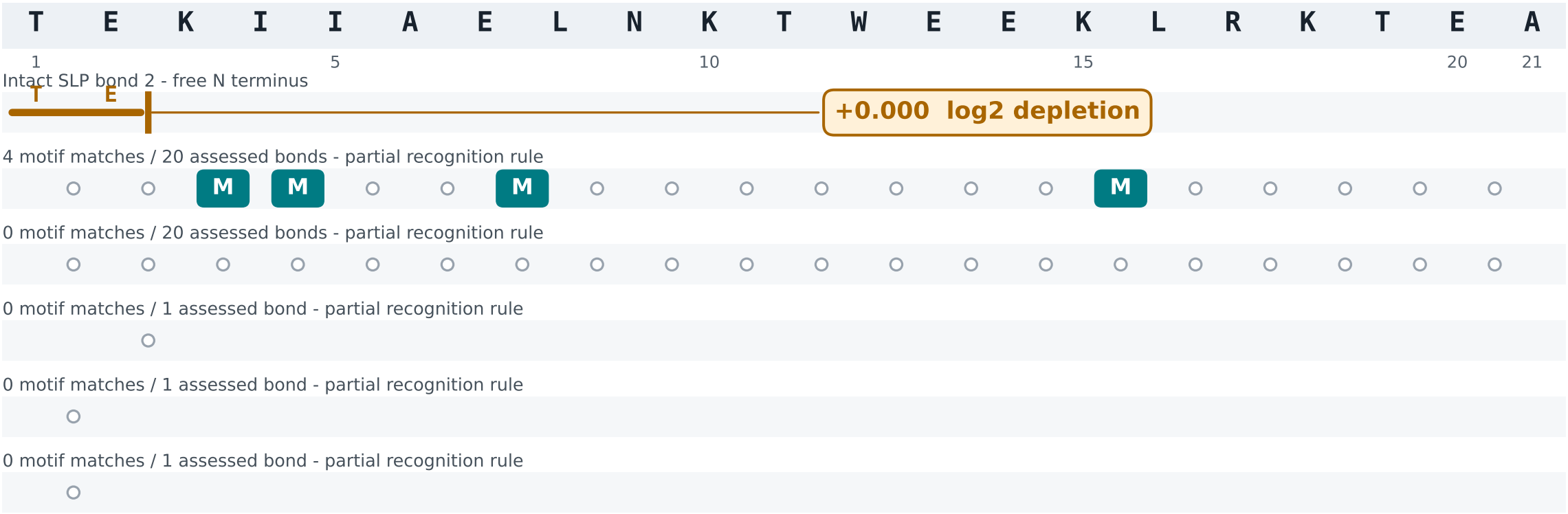
ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay

NetCleave II native 0-1



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.
DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

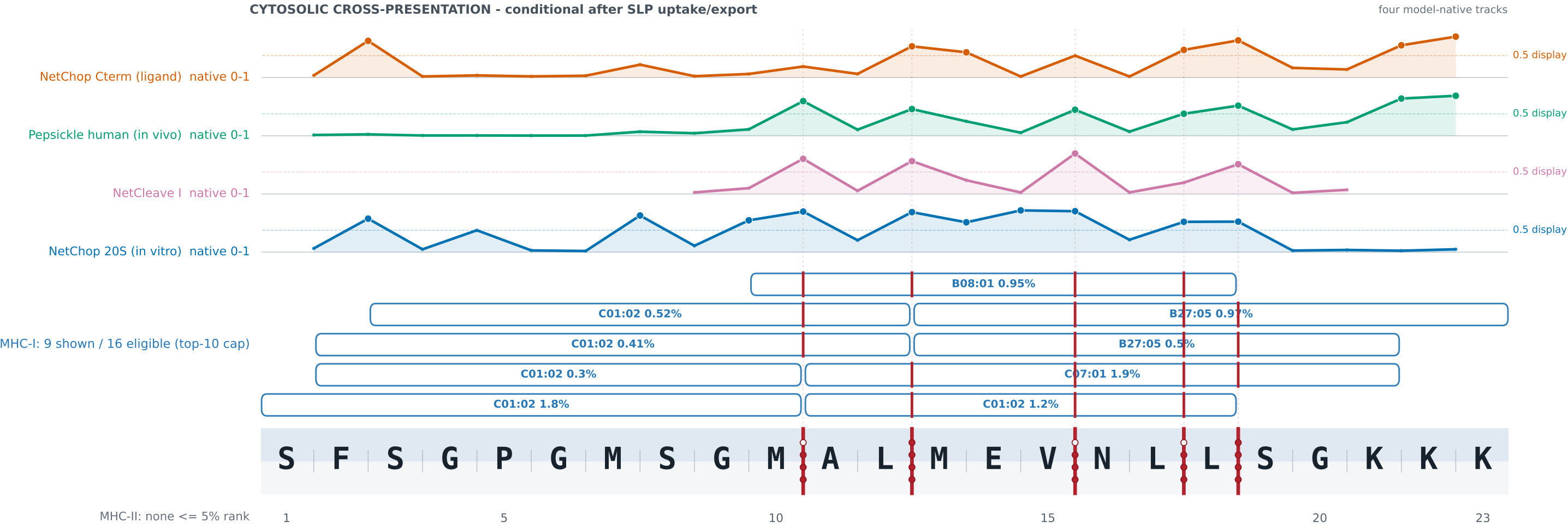
SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

CD109 p.Arg1310Met | JLF V3 | 23 aa

minimal epitope/window not disclosed for this SLP



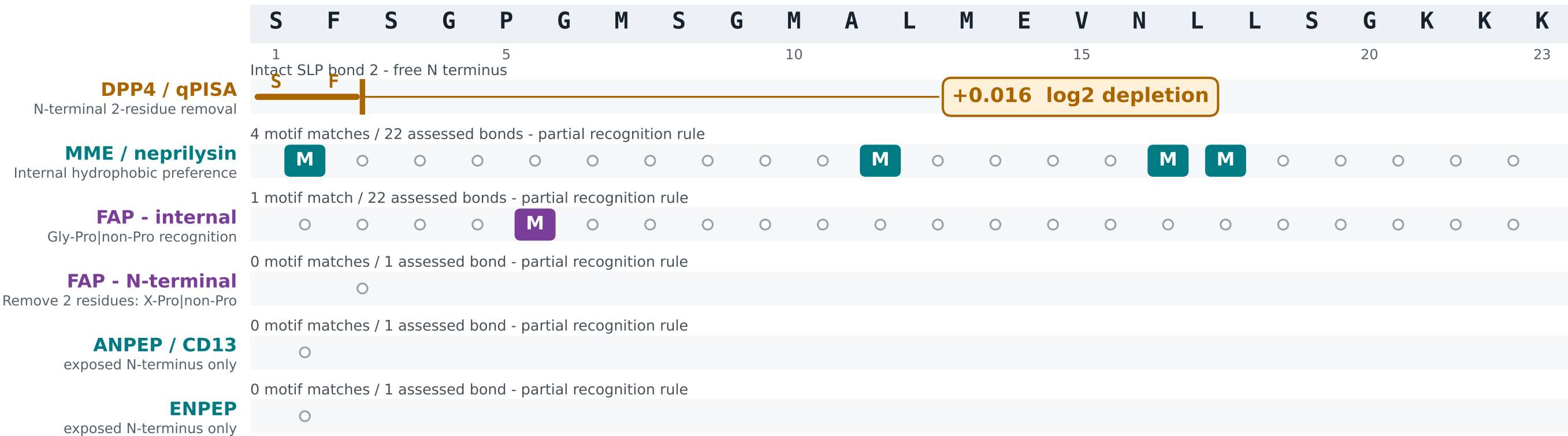
ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay

NetCleave II native 0-1

0.5 display

EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

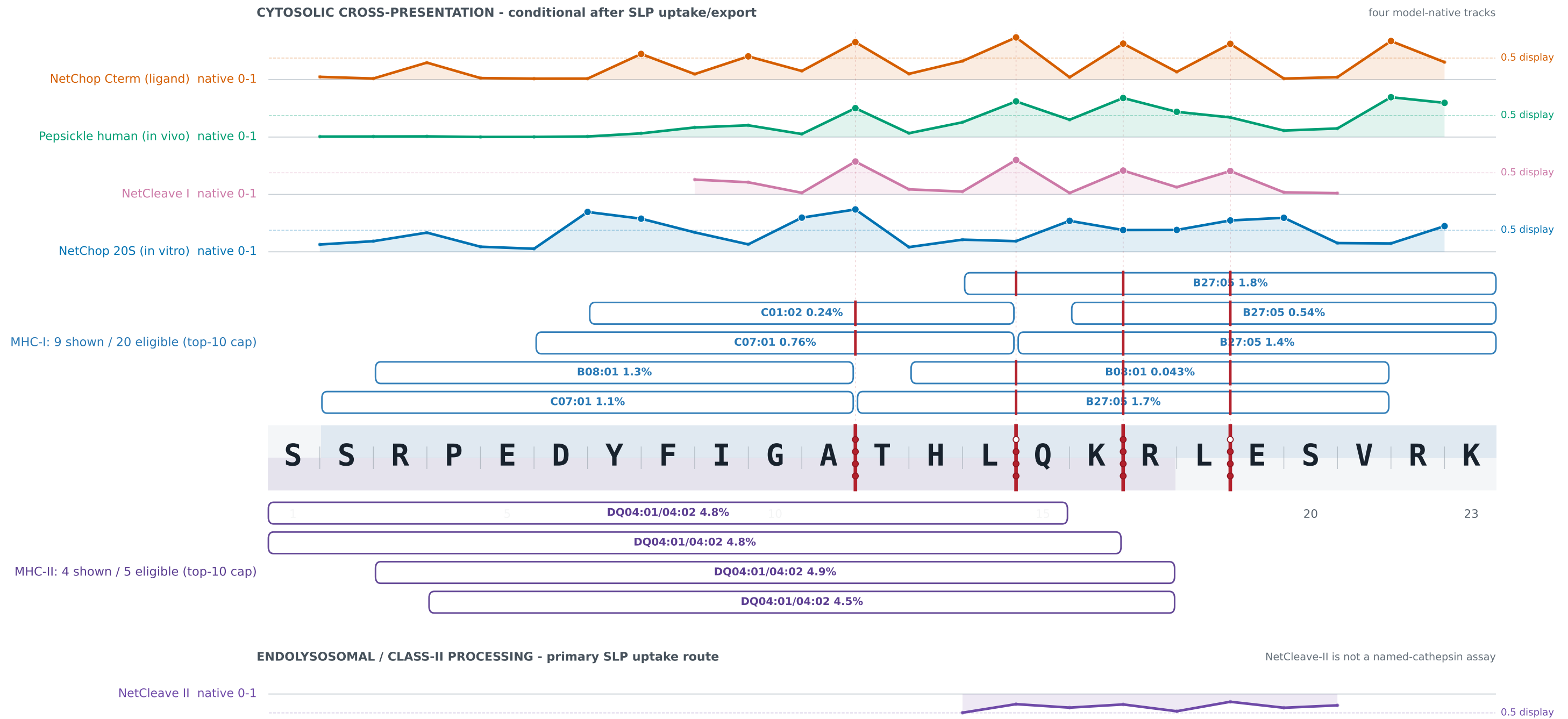
SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

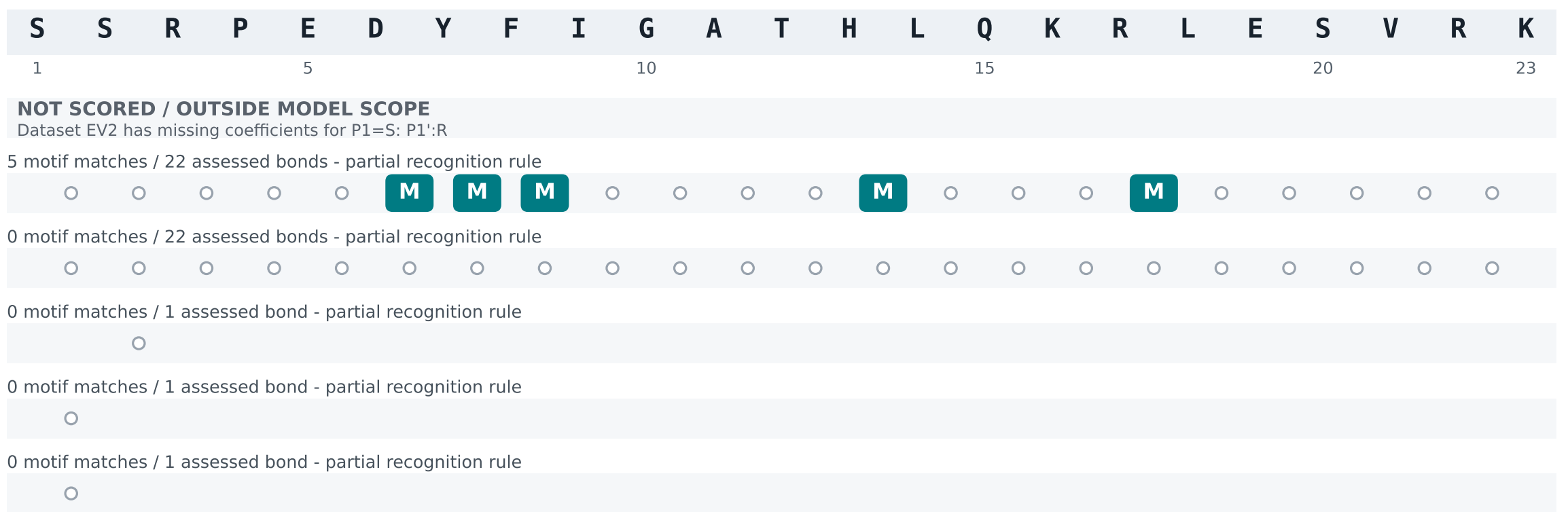
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

CENPL p.Pro26His | JLF V3 | 23 aa

minimal epitope/window not disclosed for this SLP



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and ≥ 3 reached 0.5. Four filled/open beads show model concurrence, never probability.

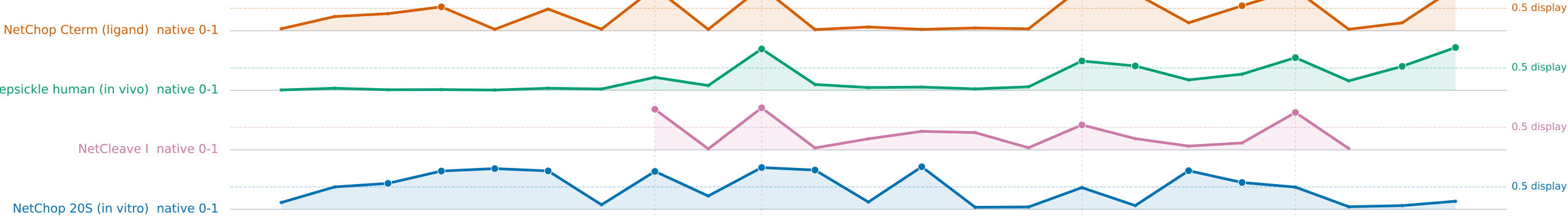
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

DYNC1H1 p.Val314Ile | JLF V3, JLF V2 | 24 aa

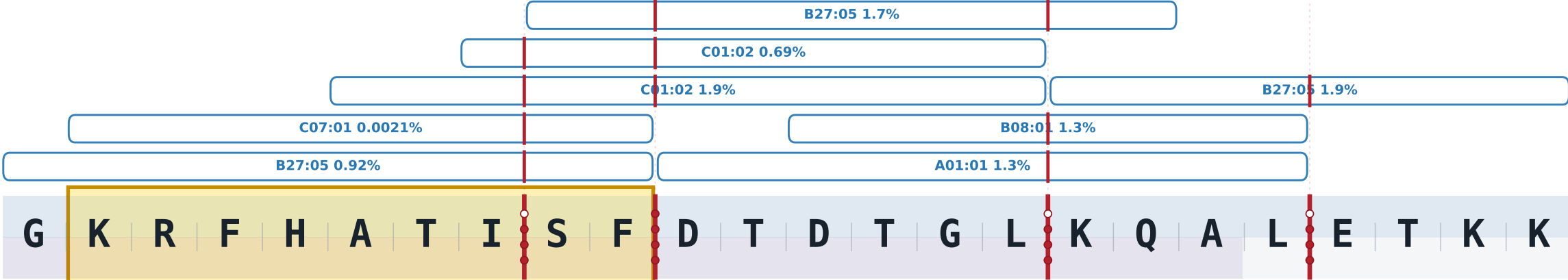
disclosed minimal epitope KRFHATISF at 2-10

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

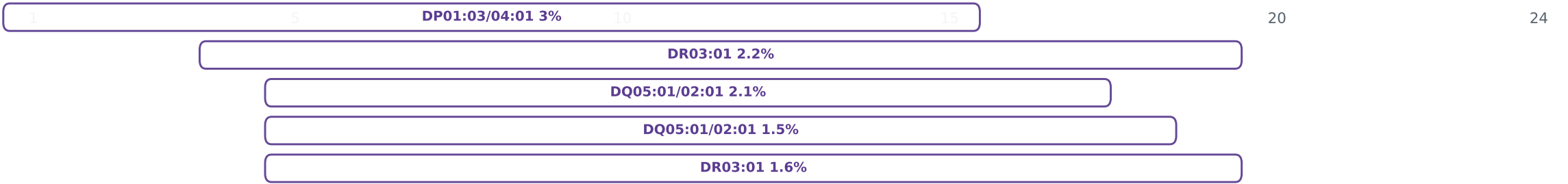
four model-native tracks



MHC-I: 8 shown / 18 eligible (top-10 cap)



MHC-II: 5 shown / 47 eligible (top-10 cap)



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



DPP4 / qPISA

N-terminal 2-residue removal

NOT SCORED / OUTSIDE MODEL SCOPE

Dataset EV2 has missing coefficients for P1=K; P1':R

5 motif matches / 23 assessed bonds - partial recognition rule

MME / neprilysin

Internal hydrophobic preference



0 motif matches / 23 assessed bonds - partial recognition rule

FAP - internal

Gly-Pro|non-Pro recognition



0 motif matches / 1 assessed bond - partial recognition rule

FAP - N-terminal

Remove 2 residues: X-Pro|non-Pro



0 motif matches / 1 assessed bond - partial recognition rule

ANPEP / CD13

exposed N-terminus only



0 motif matches / 1 assessed bond - partial recognition rule

ENPEP

exposed N-terminus only



M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

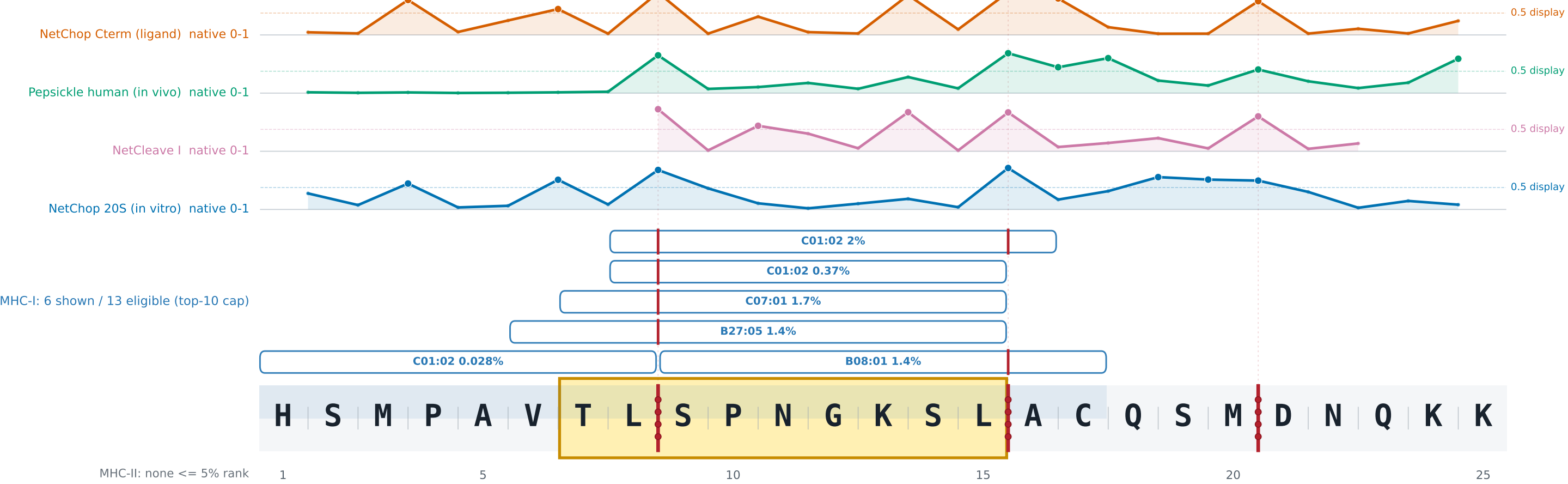
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

CDC40 p.Trp475Ser | JLF V3 | 25 aa

disclosed minimal epitope TLSPNGKSL at 7-15

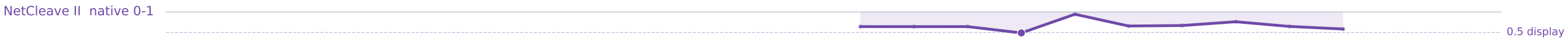
CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

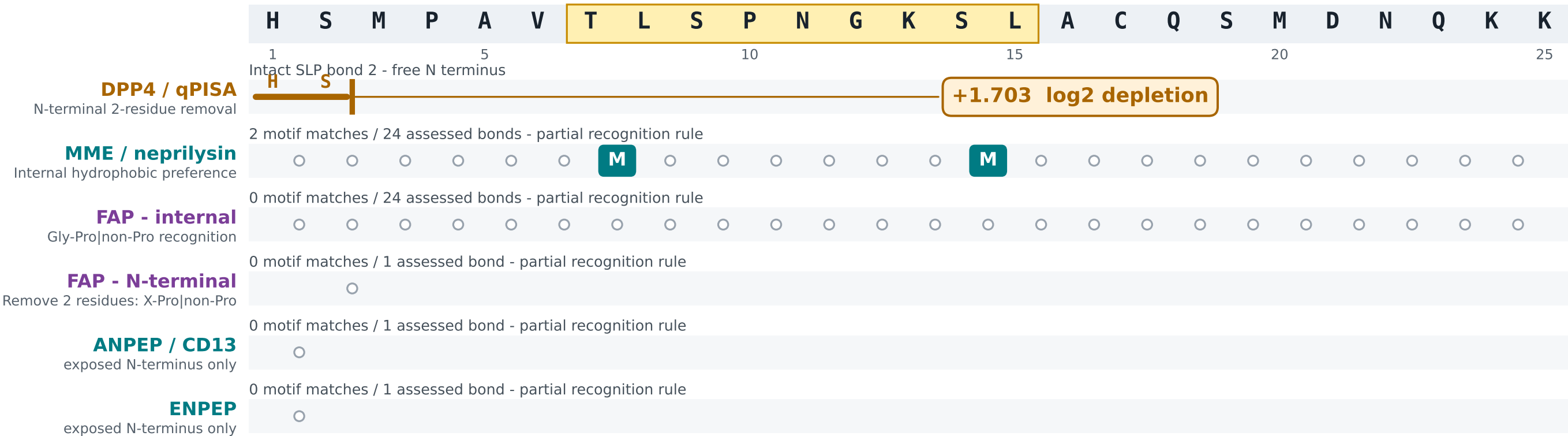


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

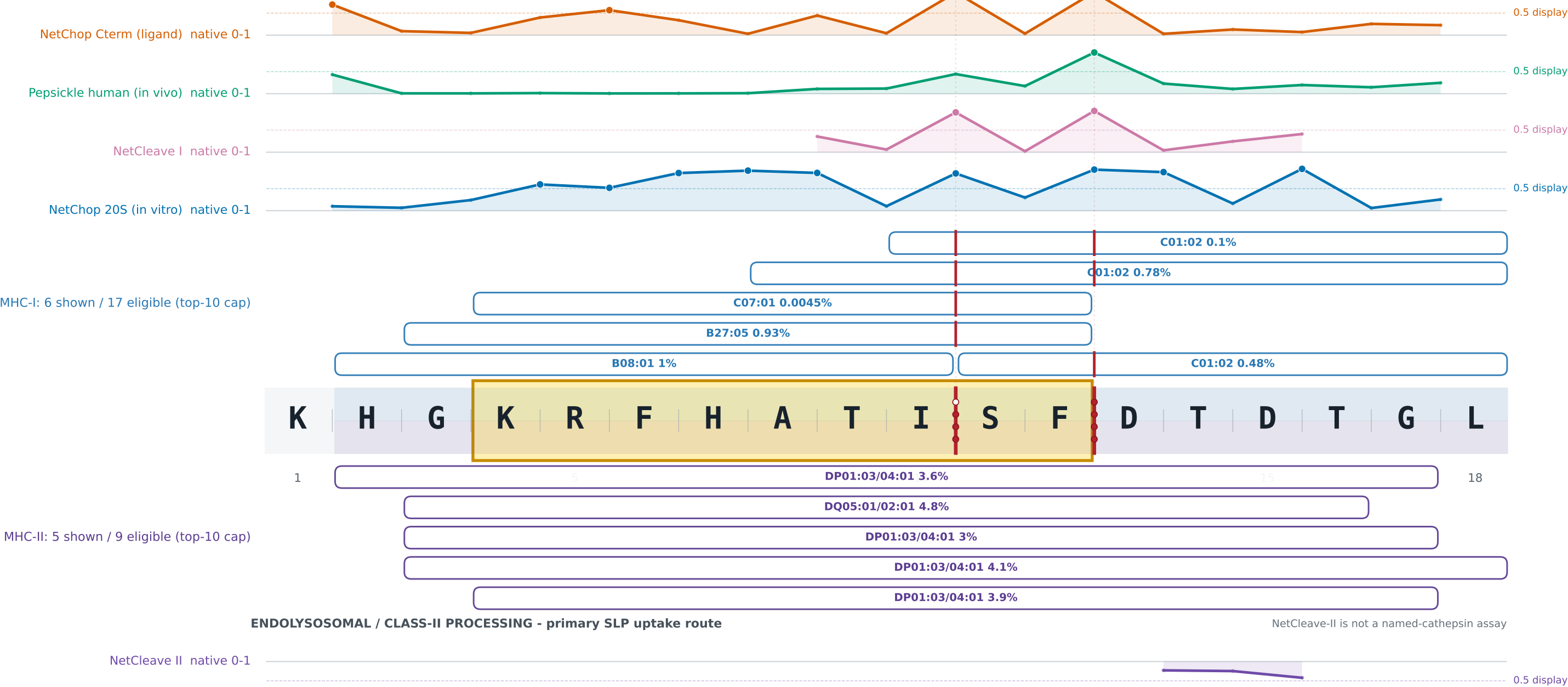
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

DYNC1H1 p.Val314Ile | CeGaT | 18 aa

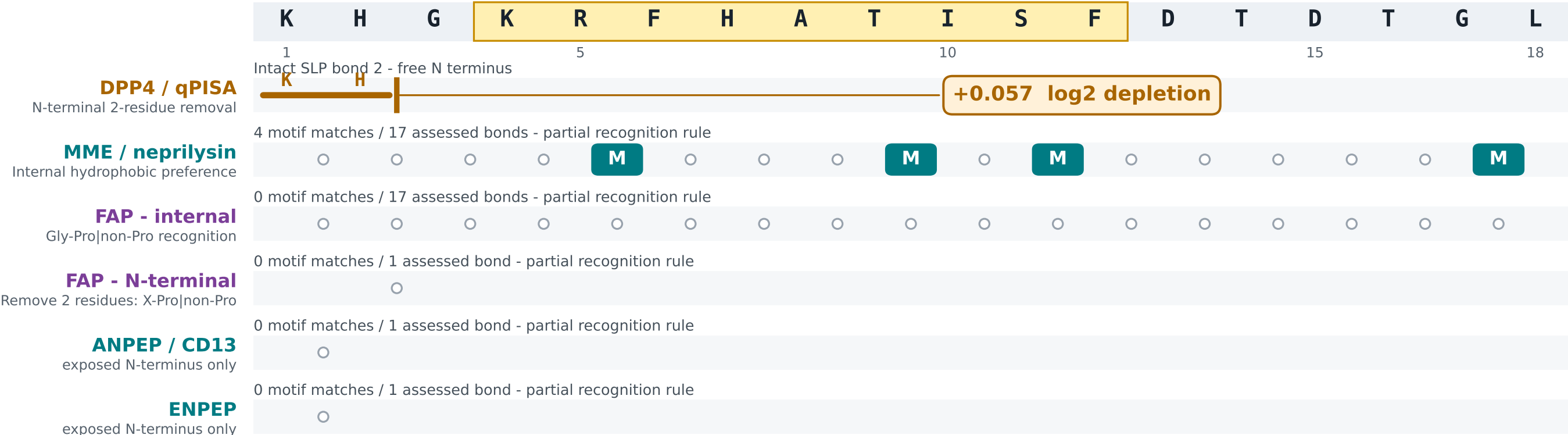
disclosed minimal epitope KRFHATISF at 4-12

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

SMC5 p.Arg371Lys | JLF V3 | 21 aa

disclosed minimal epitope RQKRIGNTR at 2-10

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

NetChop Cterm (ligand) native 0-1

Pepsickle human (in vivo) native 0-1

NetCleave I native 0-1

NetChop 20S (in vitro) native 0-1

0.5 display

0.5 display

0.5 display

0.5 display

B27:05 0.76%

C07:01 0.31%

B27:05 0.58%

B27:05 0.54%

A01:01 0.87%

B27:05 0.2%

C01:02 0.019%

B27:05 1.9%

B08:01 1.1%

D

R

Q

K

R

I

G

N

T

R

K

M

I

E

D

L

Q

N

E

L

K

1

5

DR08:01 3.2%

15

20

21

DR08:01 3.4%

DR08:01 3.7%

DR08:01 2.9%

DR03:01 3.4%

ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay

NetCleave II native 0-1

0.5 display

EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure

D

R

Q

K

R

I

G

N

T

R

K

M

I

E

D

L

Q

N

E

L

K

1

5

10

15

20

21

DPP4 / qPISA

N-terminal 2-residue removal

NOT SCORED / OUTSIDE MODEL SCOPE

Dataset EV2 has missing coefficients for P1=R: P1':Q

4 motif matches / 20 assessed bonds - partial recognition rule

MME / neprilysin

Internal hydrophobic preference

○ ○ ○ ○ M ○ ○ ○ ○ ○ M ○ ○ M ○ ○ ○ M ○

0 motif matches / 20 assessed bonds - partial recognition rule

FAP - internal

Gly-Pro|non-Pro recognition

○ ○ ○ ○ ○ ○ ○ ○ ○ ○ ○ ○ ○ ○ ○ ○ ○ ○ ○ ○

FAP - N-terminal

Remove 2 residues: X-Pro|non-Pro

○

ANPEP / CD13

exposed N-terminus only

○

1 motif match / 1 assessed bond - partial recognition rule

ENPEP

exposed N-terminus only

M

M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

CUL9 p.Asp403Tyr | JLF V2 | 23 aa

minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

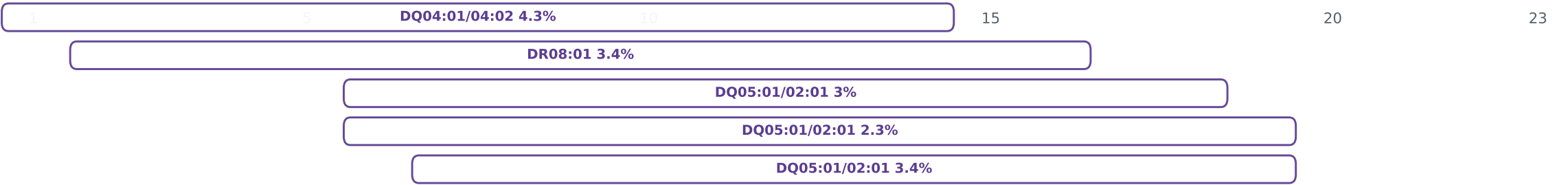
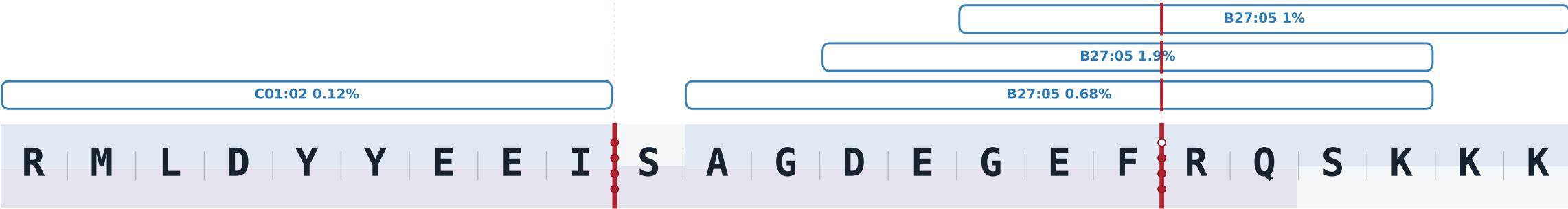
NetChop Cterm (ligand) native 0-1

Pepsickle human (in vivo) native 0-1

NetCleave I native 0-1

NetChop 20S (in vitro) native 0-1

MHC-I: 4 shown / 7 eligible (top-10 cap)



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay

NetCleave II native 0-1

EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



DPP4 / qPISA

N-terminal 2-residue removal

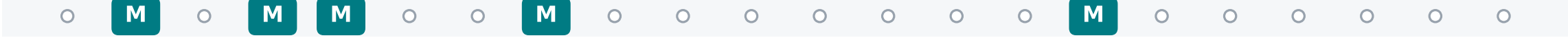
NOT SCORED / OUTSIDE MODEL SCOPE

Dataset EV2 has missing coefficients for P1=M: P2:R

5 motif matches / 22 assessed bonds - partial recognition rule

MME / neprilysin

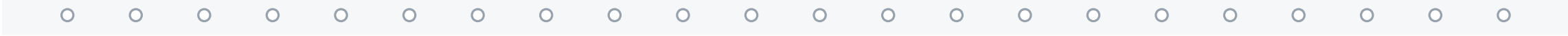
Internal hydrophobic preference



0 motif matches / 22 assessed bonds - partial recognition rule

FAP - internal

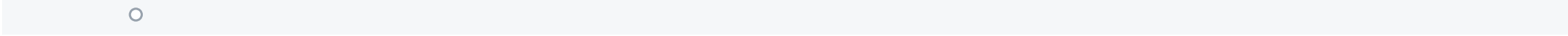
Gly-Pro|non-Pro recognition



0 motif matches / 1 assessed bond - partial recognition rule

FAP - N-terminal

Remove 2 residues: X-Pro|non-Pro



0 motif matches / 1 assessed bond - partial recognition rule

ANPEP / CD13

exposed N-terminus only



0 motif matches / 1 assessed bond - partial recognition rule

ENPEP

exposed N-terminus only



M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

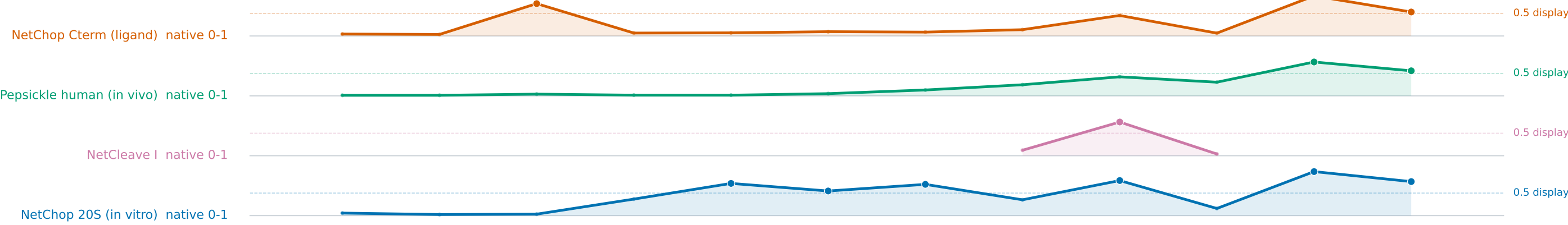
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

NME1 p.Gly24Arg | CeGaT | 13 aa

minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



MHC-I: 1 shown / 1 eligible (top-10 cap)

C01:02 1.1%



MHC-II: none <= 5% rank

1 5 10 13

ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay

NetCleave II native 0-1

0.5 display

EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure

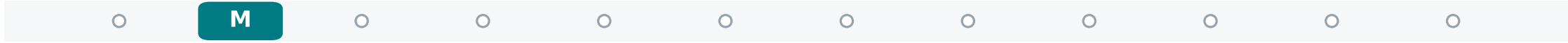


DPP4 / qPISA

N-terminal 2-residue removal

NOT SCORED / OUTSIDE MODEL SCOPE
Dataset EV2 has missing coefficients for P1=P: P2:P

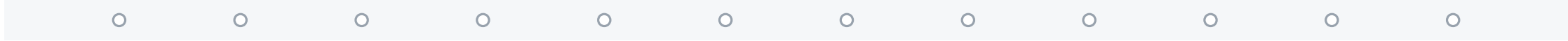
1 motif match / 12 assessed bonds - partial recognition rule



MME / neprilysin

Internal hydrophobic preference

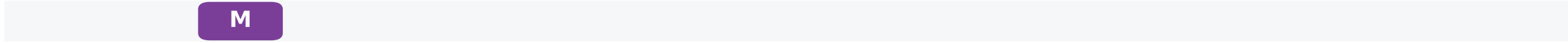
0 motif matches / 12 assessed bonds - partial recognition rule



FAP - internal

Gly-Pro|non-Pro recognition

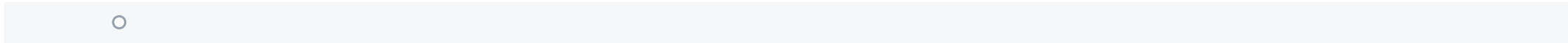
1 motif match / 1 assessed bond - partial recognition rule



FAP - N-terminal

Remove 2 residues: X-Pro|non-Pro

0 motif matches / 1 assessed bond - partial recognition rule



ANPEP / CD13

exposed N-terminus only

0 motif matches / 1 assessed bond - partial recognition rule



ENPEP

exposed N-terminus only

M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

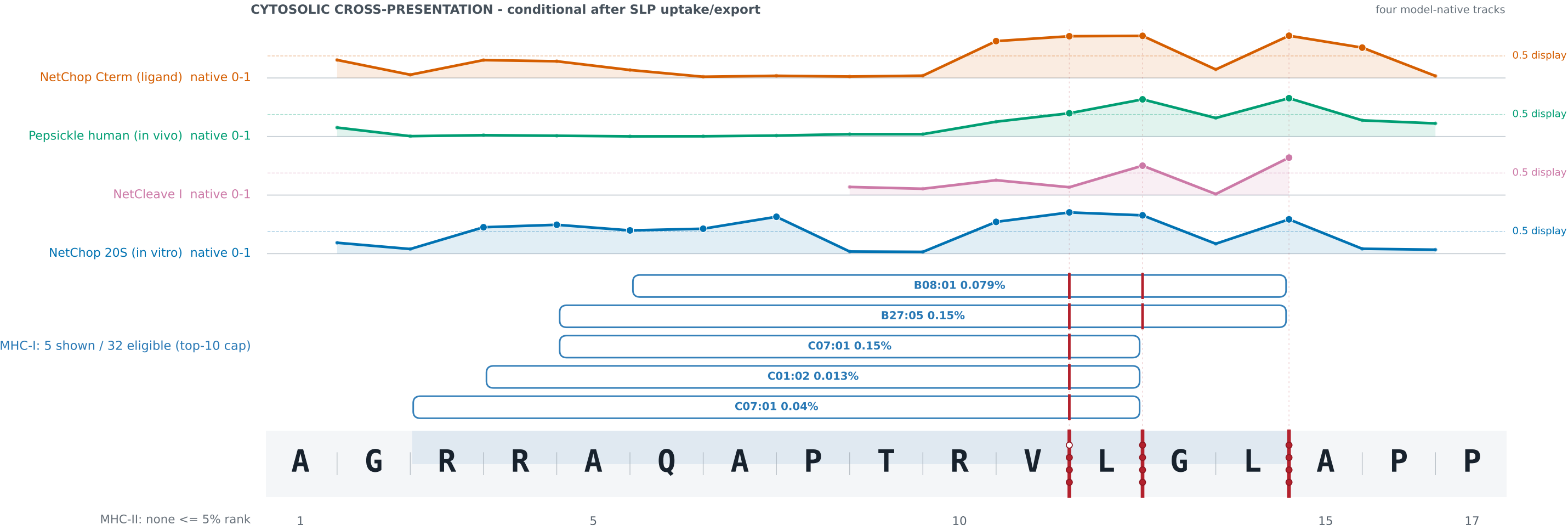
SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

MYO15B fs | JLF V2, JLF V1 | 17 aa

minimal epitope/window not disclosed for this SLP



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure

	A	G	R	R	A	Q	A	P	T	R	V	L	G	L	A	P	P
	1				5					10					15		17
DPP4 / qPISA N-terminal 2-residue removal	NOT SCORED / OUTSIDE MODEL SCOPE Dataset EV2 has missing coefficients for P1=G; P1':R																
	2 motif matches / 16 assessed bonds - partial recognition rule																
	○	○	○	○	○	○	○	○	○	○	M	○	M	○	○	○	
MME / neprilysin Internal hydrophobic preference	0 motif matches / 16 assessed bonds - partial recognition rule																
	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	○	
FAP - internal Gly-Pro non-Pro recognition	0 motif matches / 1 assessed bond - partial recognition rule																
	○																
FAP - N-terminal Remove 2 residues: X-Pro non-Pro	1 motif match / 1 assessed bond - partial recognition rule																
	M																
ANPEP / CD13 exposed N-terminus only	0 motif matches / 1 assessed bond - partial recognition rule																
	○																
ENPEP exposed N-terminus only																	

M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

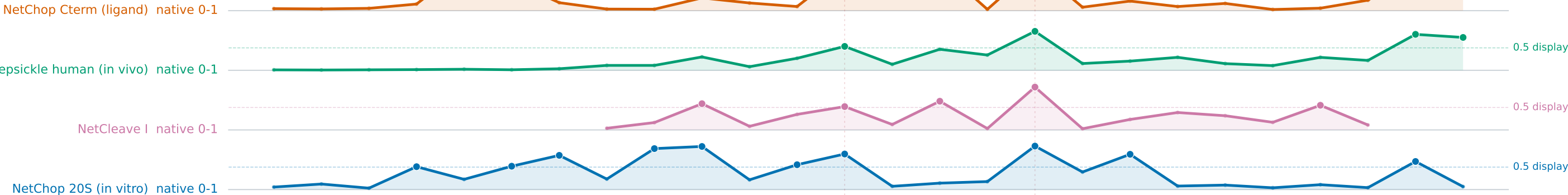
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

VPS72 p.Lys115Arg | JLF V3 | 27 aa

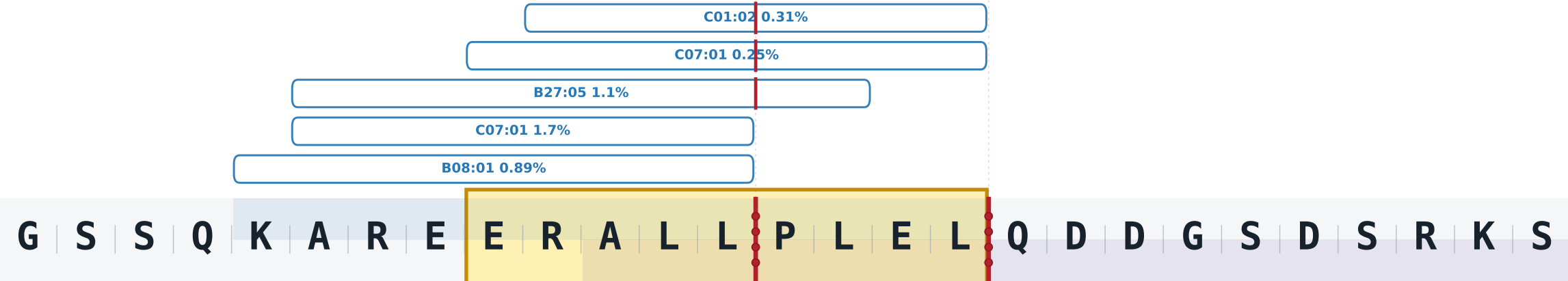
disclosed minimal epitope ERALLPLEL at 9-17

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

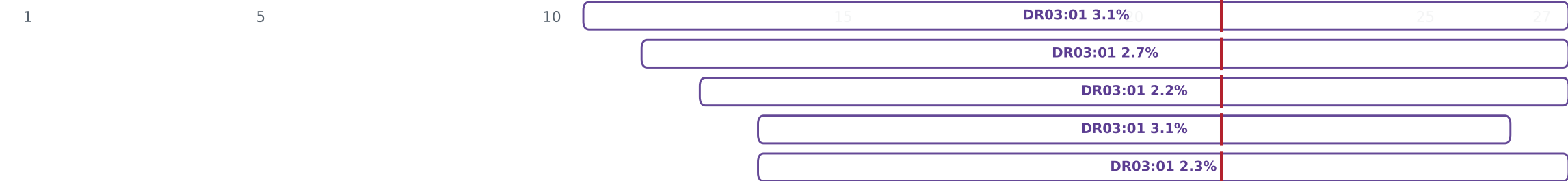
four model-native tracks



MHC-I: 5 shown / 12 eligible (top-10 cap)



MHC-II: 5 shown / 10 eligible (top-10 cap)

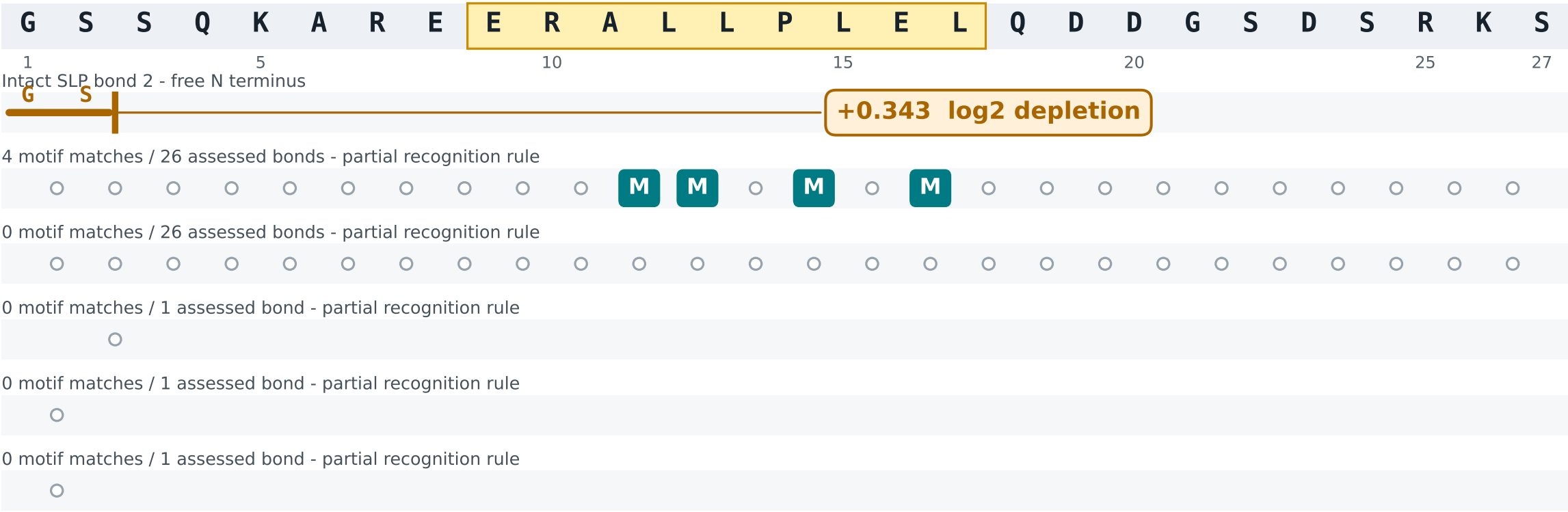


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.
DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

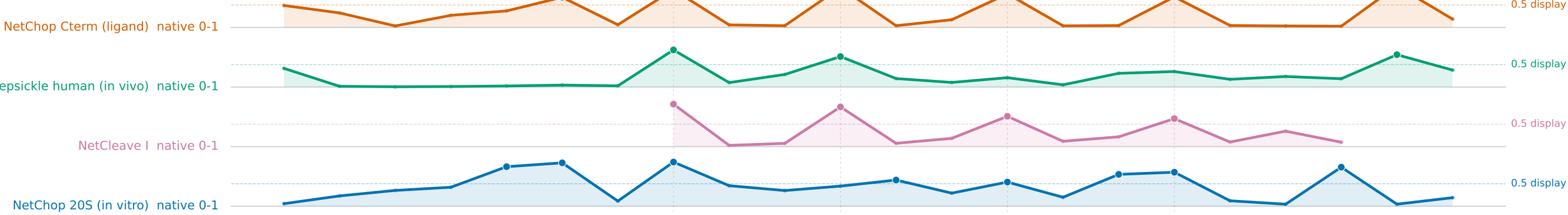
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

EXOC4 p.Ser34Ile | JLF V3, JLF V2 | 23 aa

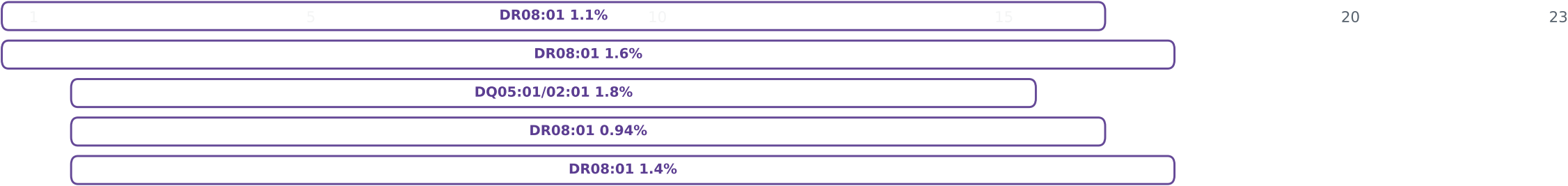
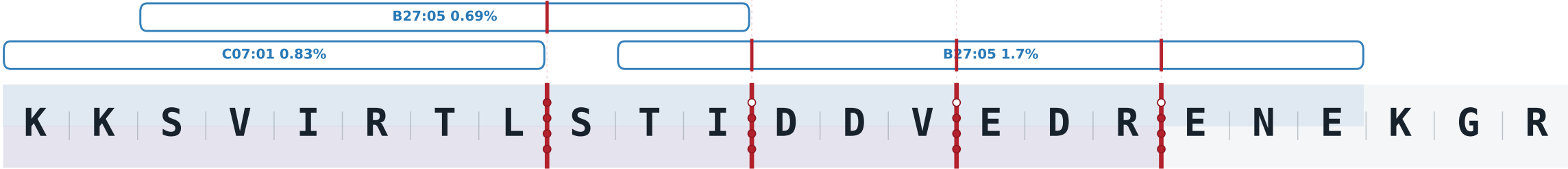
minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



MHC-I: 3 shown / 4 eligible (top-10 cap)



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



DPP4 / qPISA

N-terminal 2-residue removal

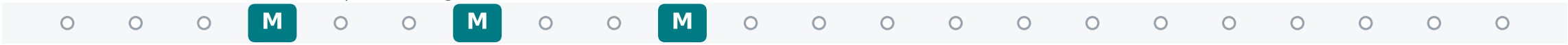
NOT SCORED / OUTSIDE MODEL SCOPE

Dataset EV2 has missing coefficients for P1=K: P1':S

3 motif matches / 22 assessed bonds - partial recognition rule

MME / neprilysin

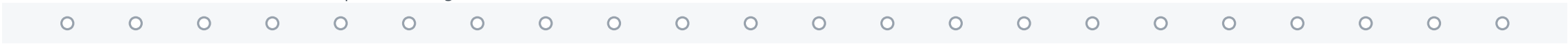
Internal hydrophobic preference



0 motif matches / 22 assessed bonds - partial recognition rule

FAP - internal

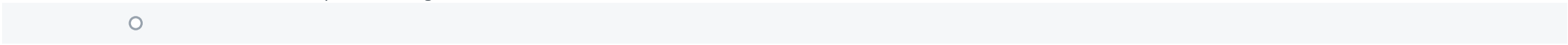
Gly-Pro|non-Pro recognition



0 motif matches / 1 assessed bond - partial recognition rule

FAP - N-terminal

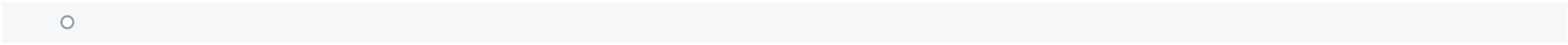
Remove 2 residues: X-Pro|non-Pro



0 motif matches / 1 assessed bond - partial recognition rule

ANPEP / CD13

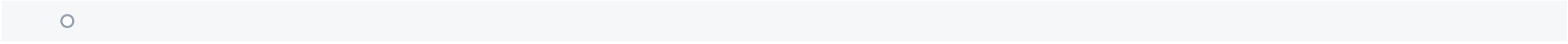
exposed N-terminus only



0 motif matches / 1 assessed bond - partial recognition rule

ENPEP

exposed N-terminus only



M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

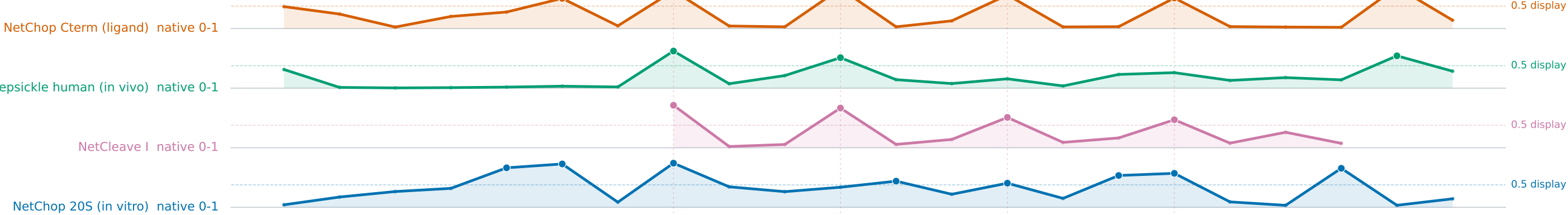
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

EPG5 p.Ala869Thr | CeGaT | 23 aa

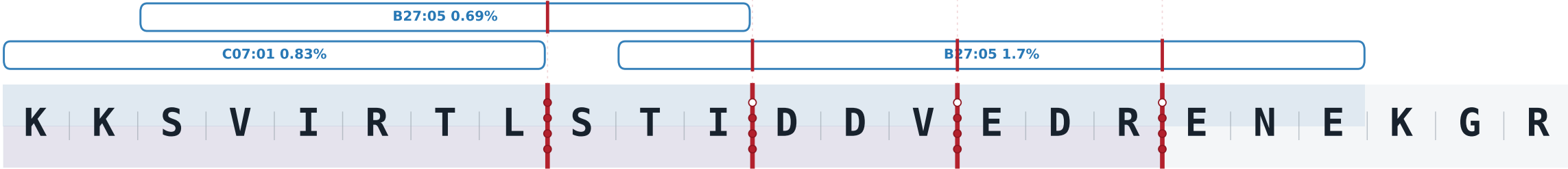
minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

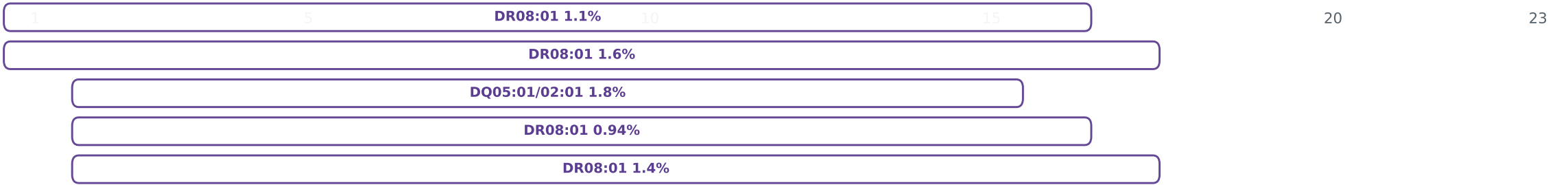
four model-native tracks



MHC-I: 3 shown / 4 eligible (top-10 cap)



MHC-II: 5 shown / 38 eligible (top-10 cap)



ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



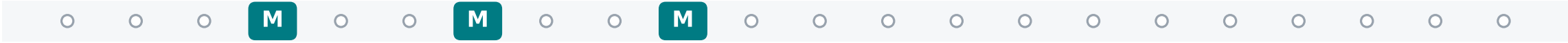
DPP4 / qPISA

N-terminal 2-residue removal

NOT SCORED / OUTSIDE MODEL SCOPE

Dataset EV2 has missing coefficients for P1=K: P1':S

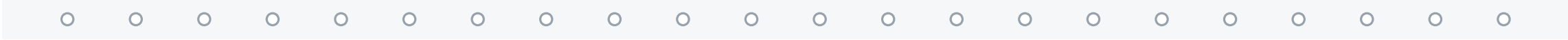
3 motif matches / 22 assessed bonds - partial recognition rule



MME / neprilysin

Internal hydrophobic preference

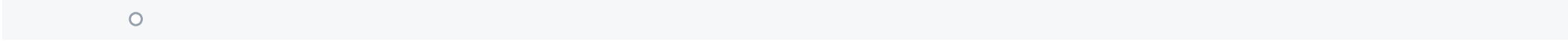
0 motif matches / 22 assessed bonds - partial recognition rule



FAP - internal

Gly-Pro|non-Pro recognition

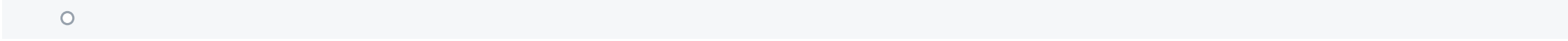
0 motif matches / 1 assessed bond - partial recognition rule



FAP - N-terminal

Remove 2 residues: X-Pro|non-Pro

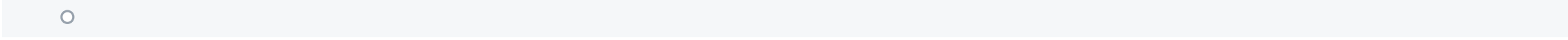
0 motif matches / 1 assessed bond - partial recognition rule



ANPEP / CD13

exposed N-terminus only

0 motif matches / 1 assessed bond - partial recognition rule



ENPEP

exposed N-terminus only

M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

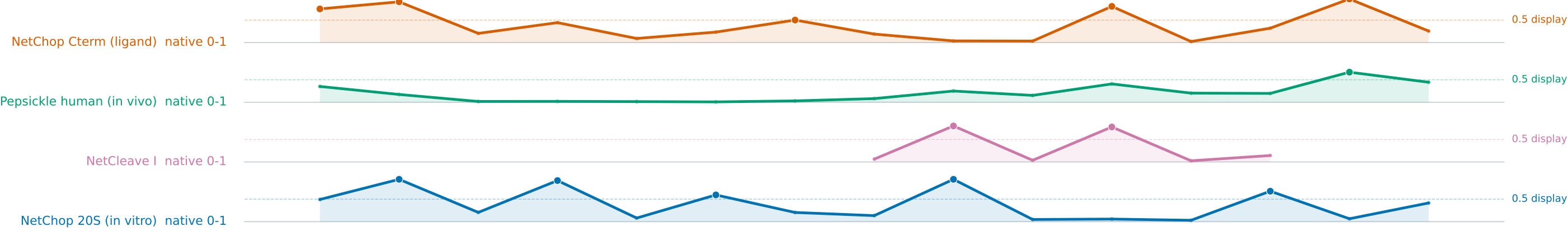
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

TECPR1 p.Thr259fs | CeGaT | 16 aa

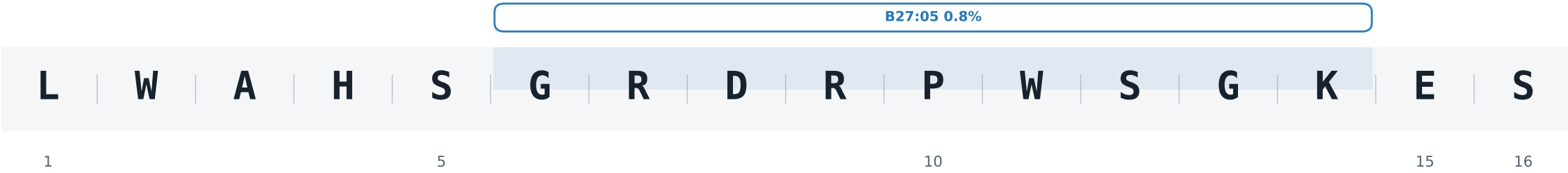
minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



MHC-II: none <= 5% rank

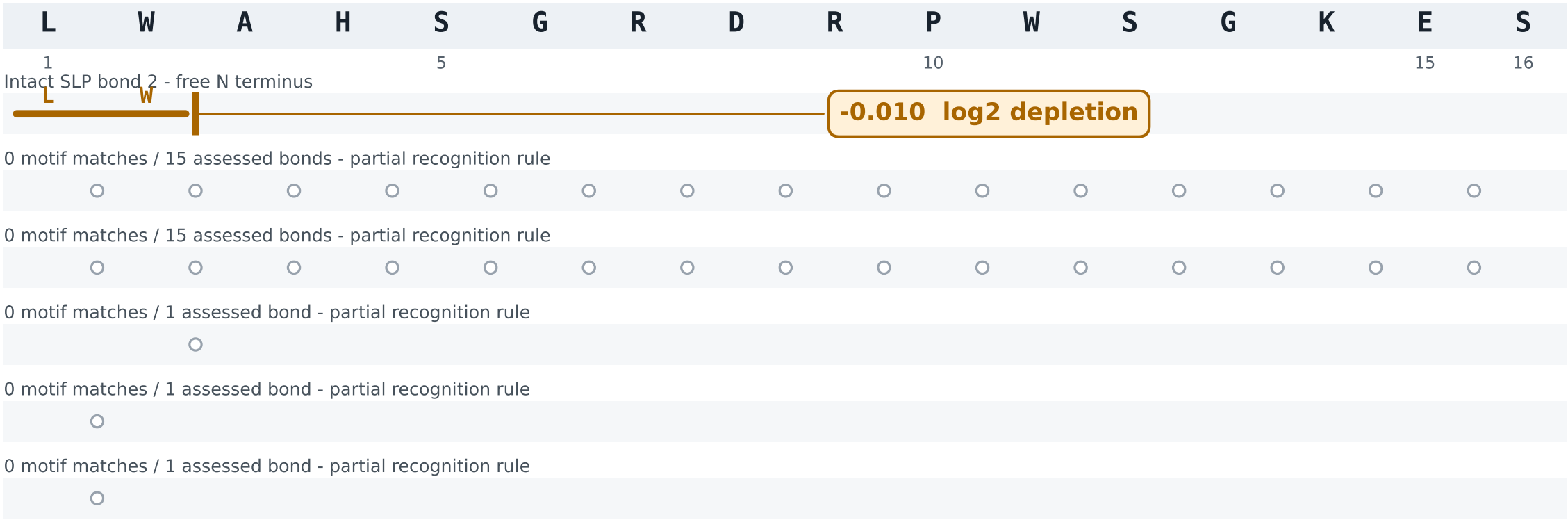


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

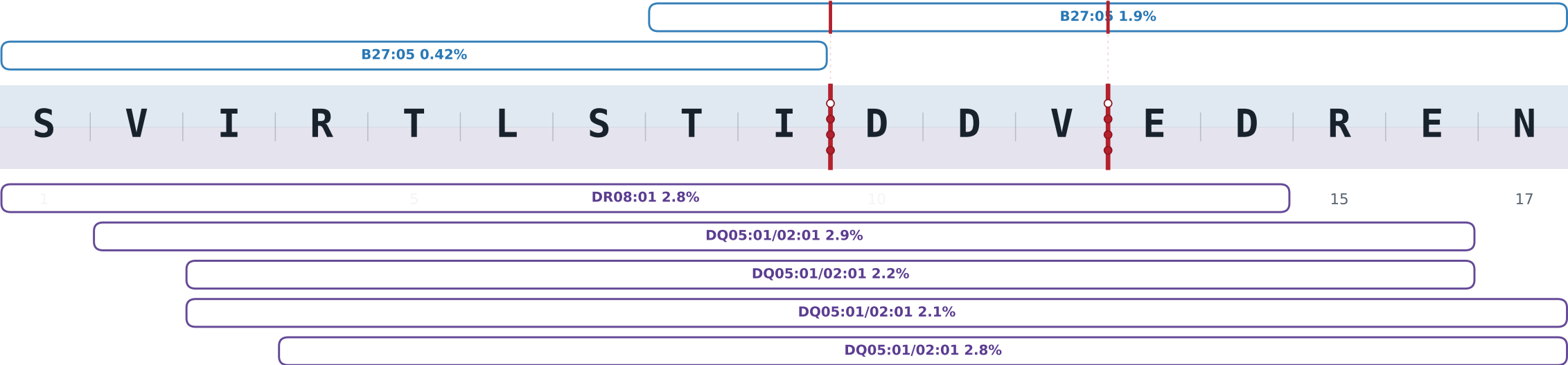
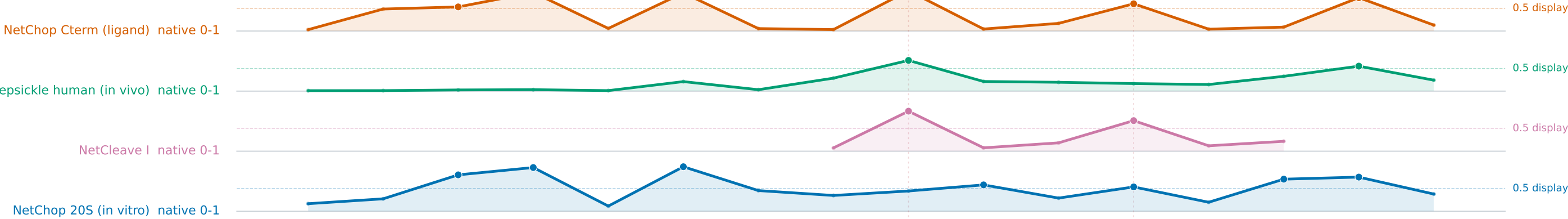
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

EXOC4 p.Ser34Ile | CeGaT | 17 aa

minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

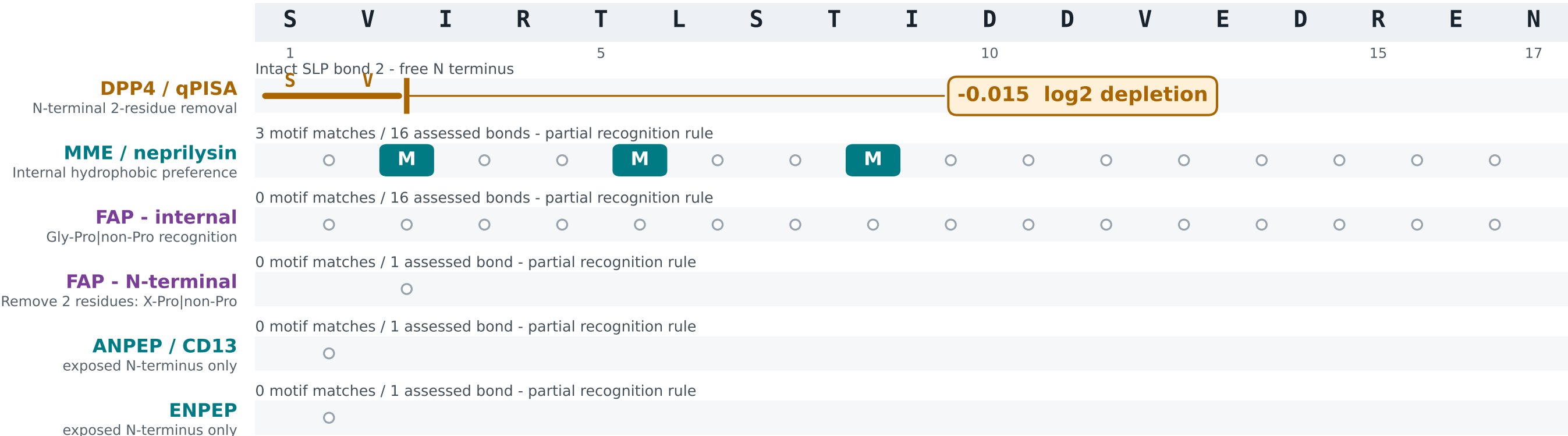


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.
DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

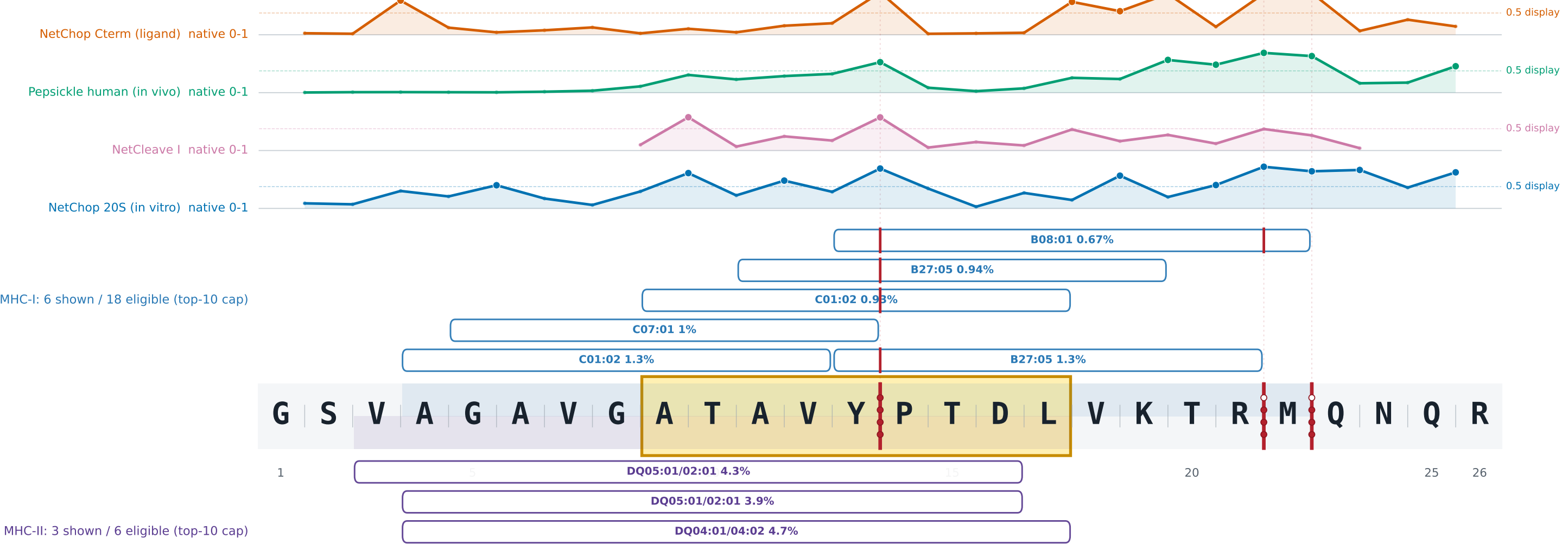
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

SLC25A12 p.Ile347Thr | JLF V2, JLF V1 | 26 aa

disclosed minimal epitope ATAVYPTDL at 9-17

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



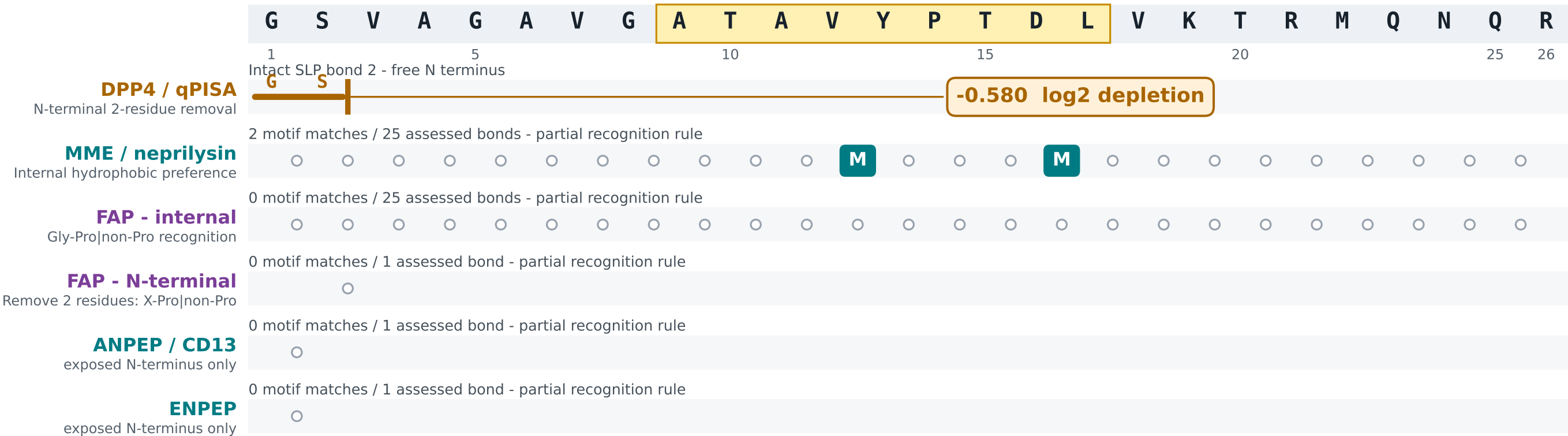
ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay

NetCleave II native 0-1

0.5 display

EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

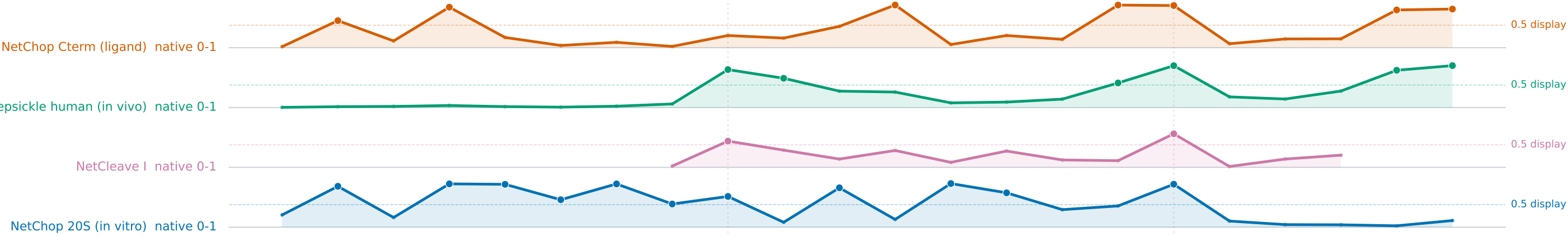
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

PTH1R p.Ile237Phe | JLF V3 | 23 aa

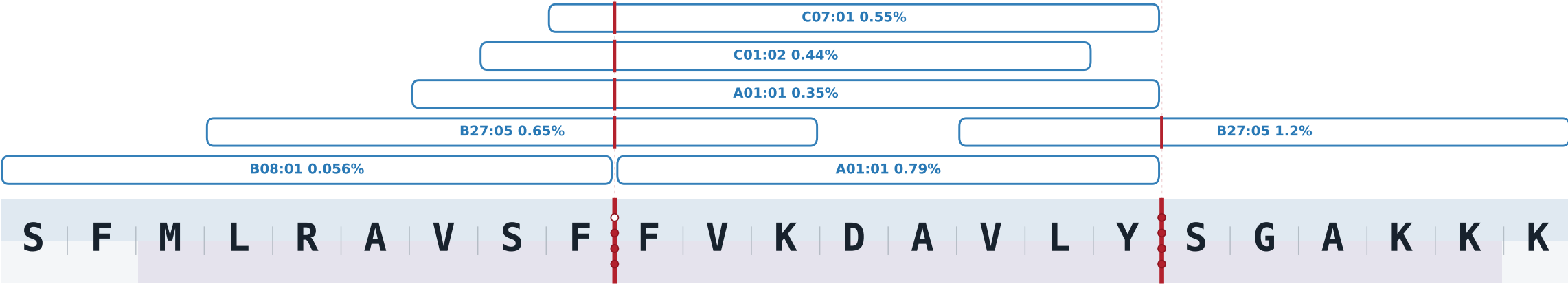
minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

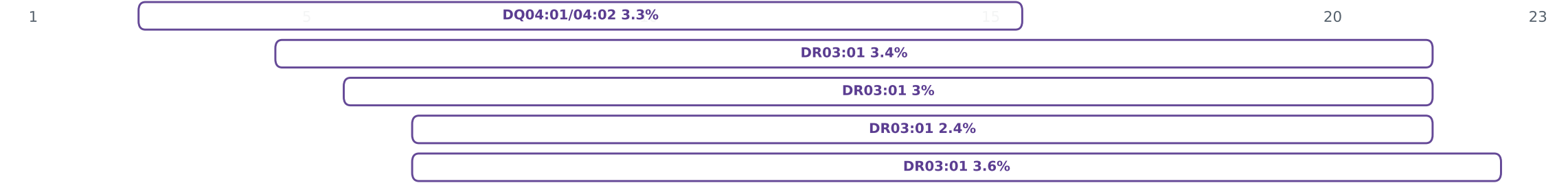
four model-native tracks



MHC-I: 7 shown / 23 eligible (top-10 cap)



MHC-II: 5 shown / 15 eligible (top-10 cap)

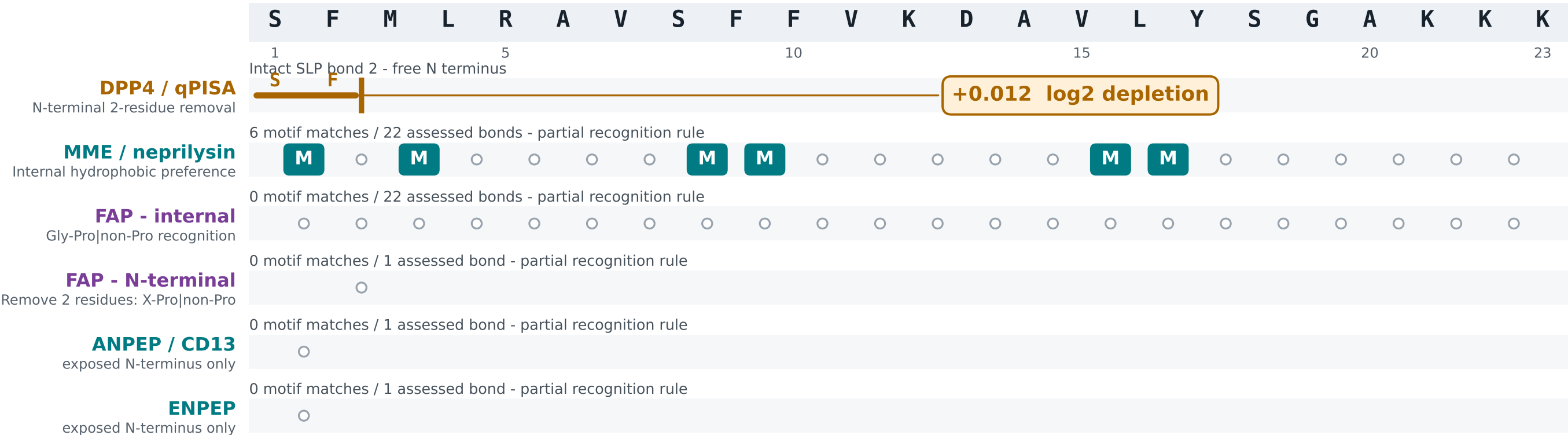


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.
DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

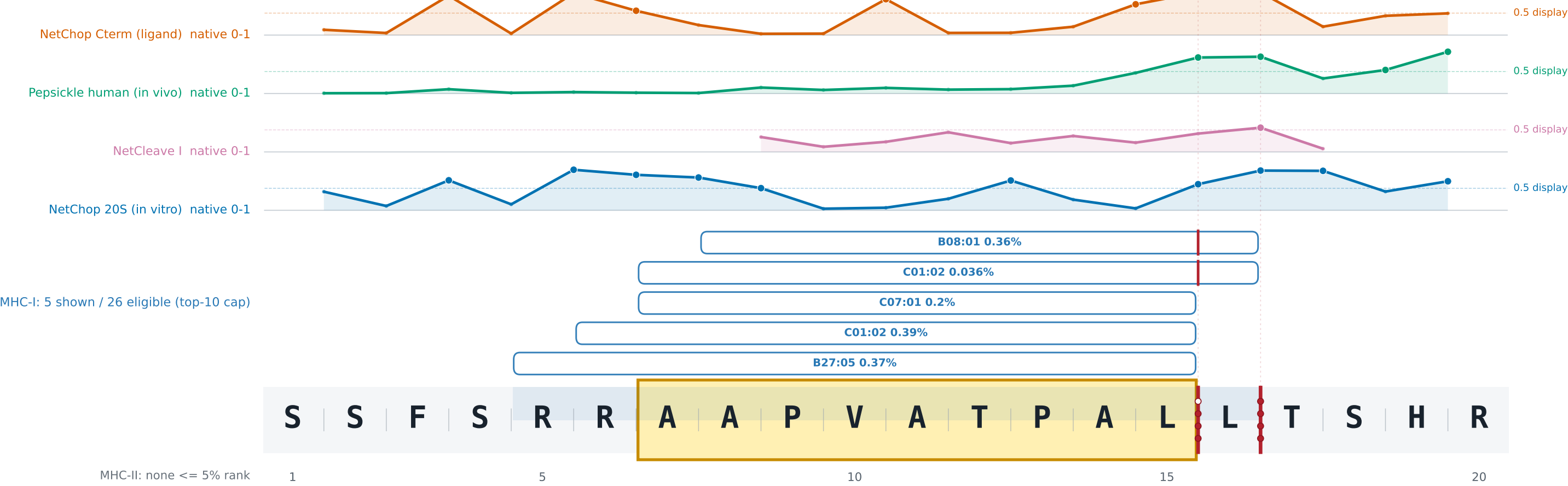
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

PIP5K1A p.Gly474fs | JLF V3 | 20 aa

disclosed minimal epitope AAPVATPAL at 7-15

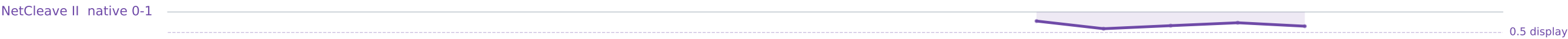
CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

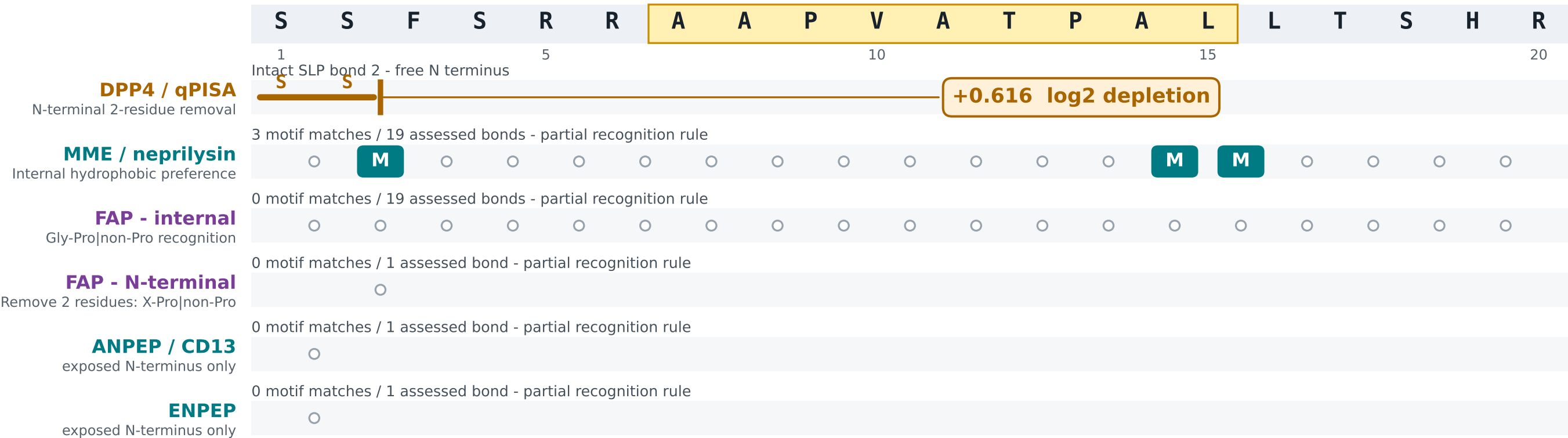


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

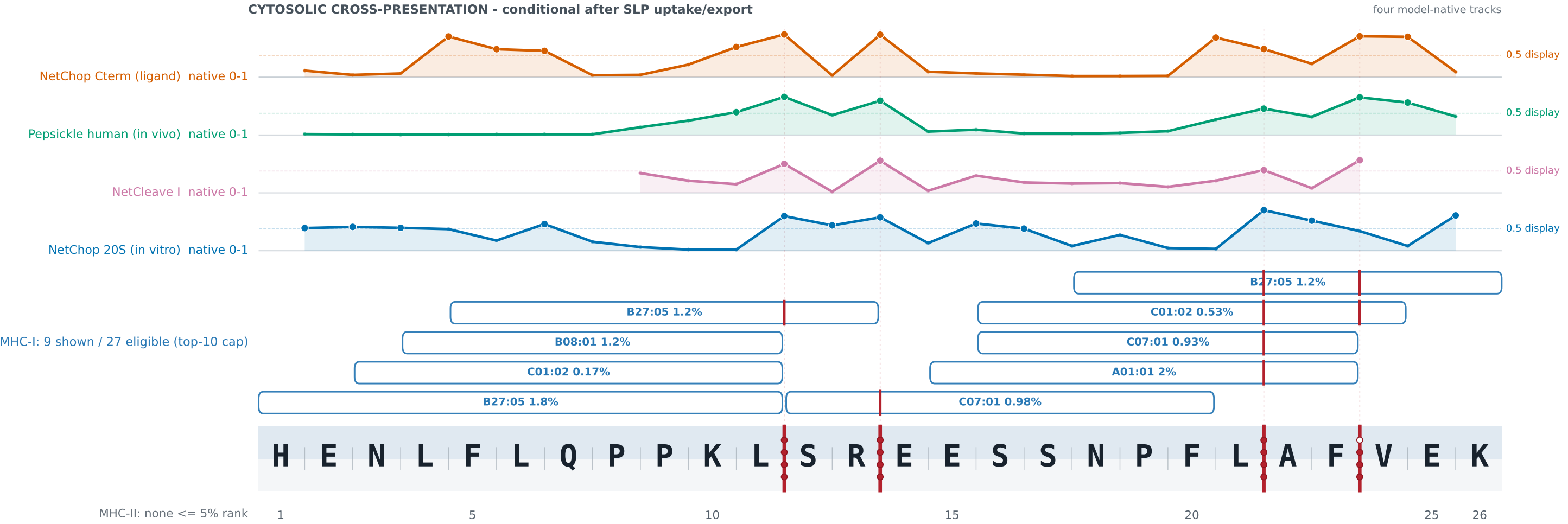
SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

KDM3B p.Pro633Ser | JLF V2, JLF V1 | 26 aa

minimal epitope/window not disclosed for this SLP

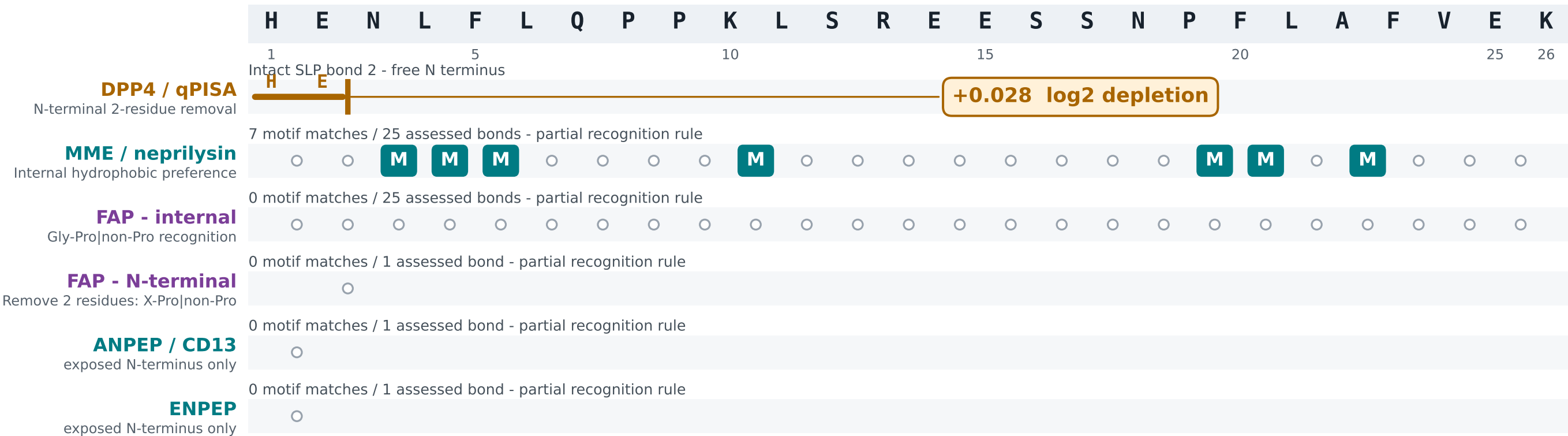


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

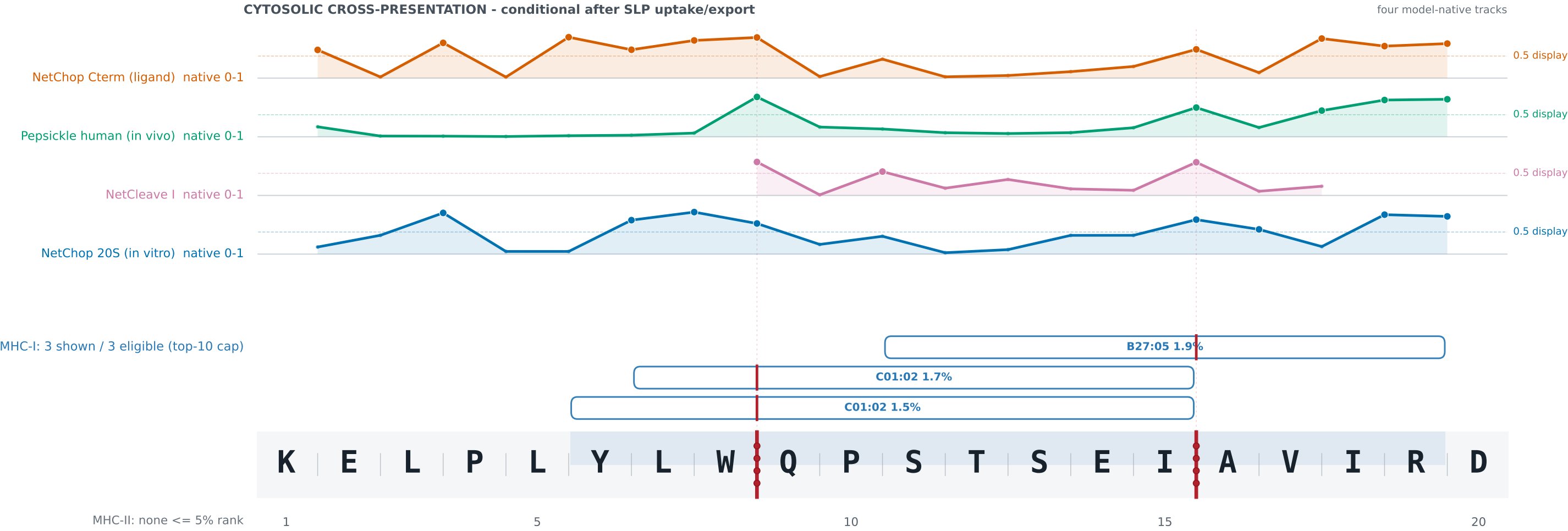
SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

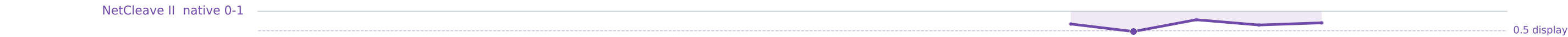
EPG5 p.Ala869Thr | JLF V1 | 20 aa

minimal epitope/window not disclosed for this SLP

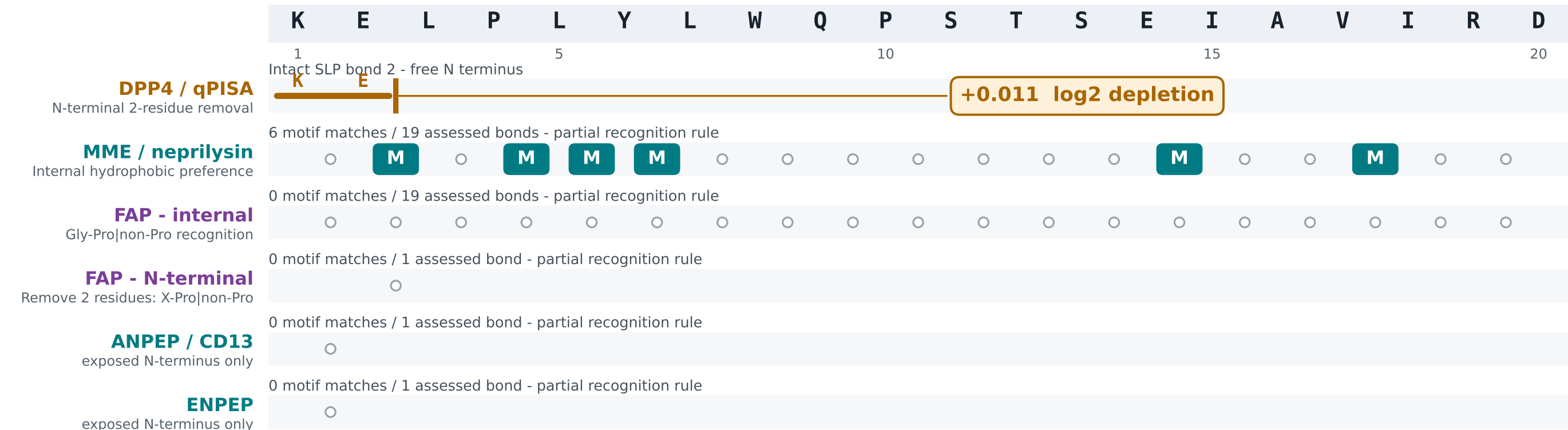


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

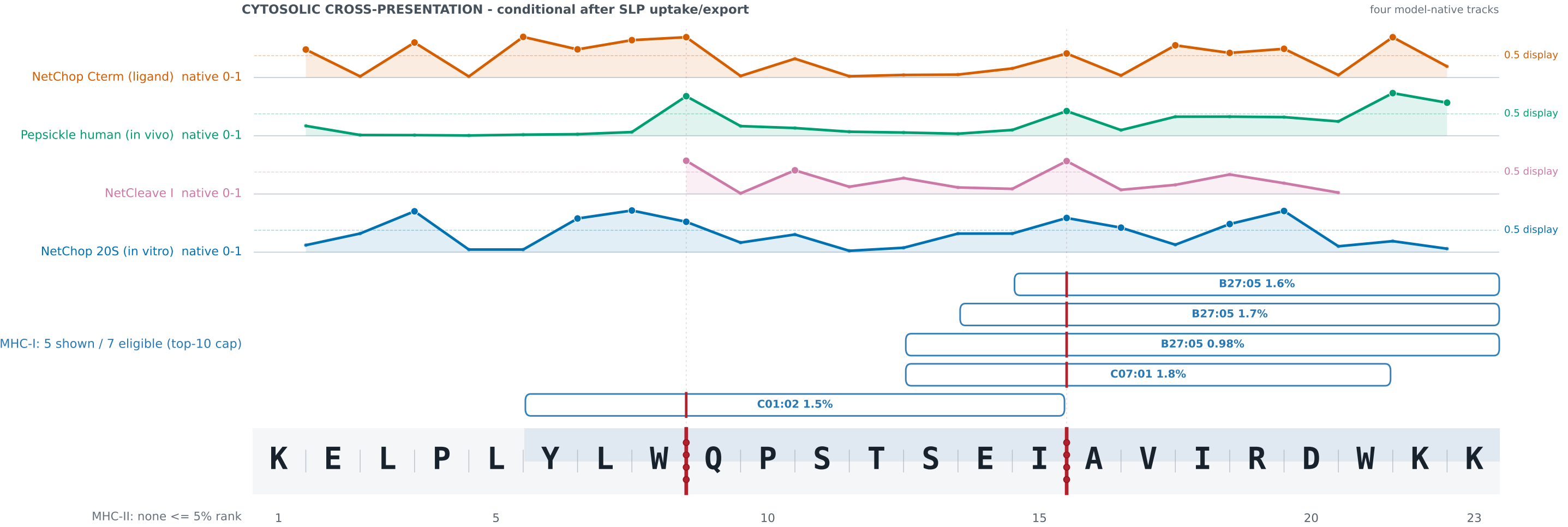
SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

EPG5 p.Ala869Thr | JLF V2 | 23 aa

minimal epitope/window not disclosed for this SLP

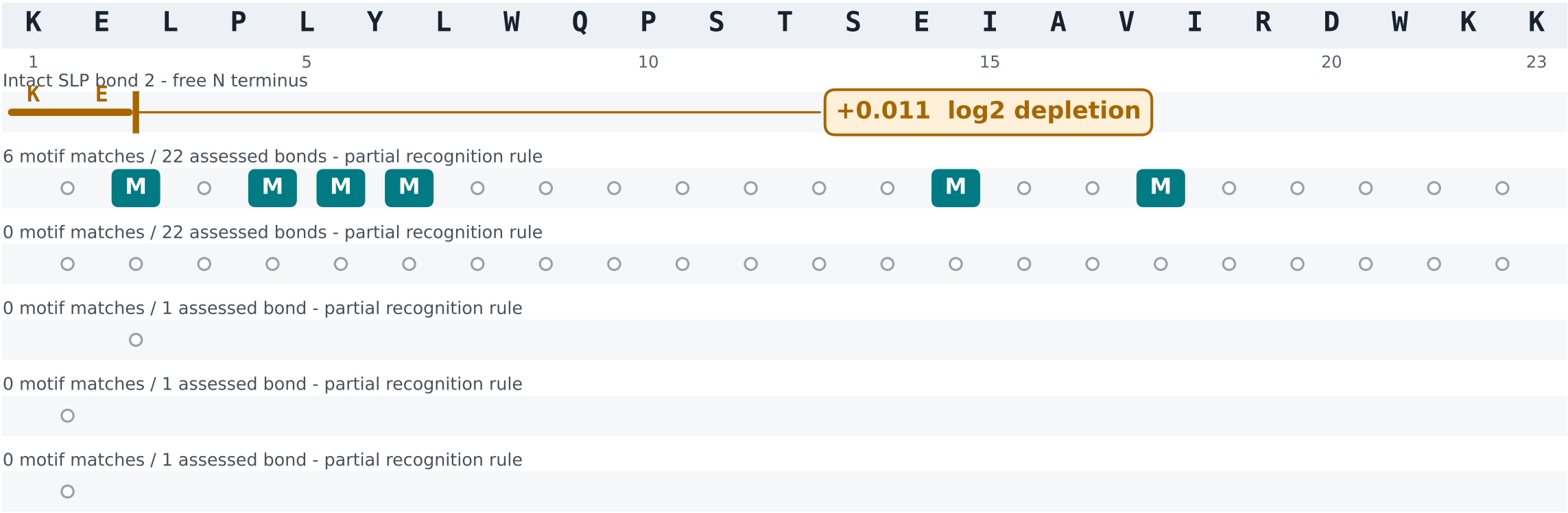


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

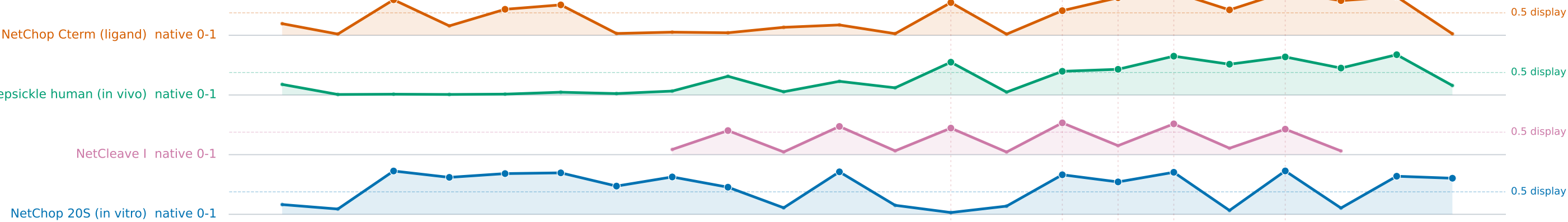
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

MAP2 p.Leu867fs | JLF V3 | 23 aa

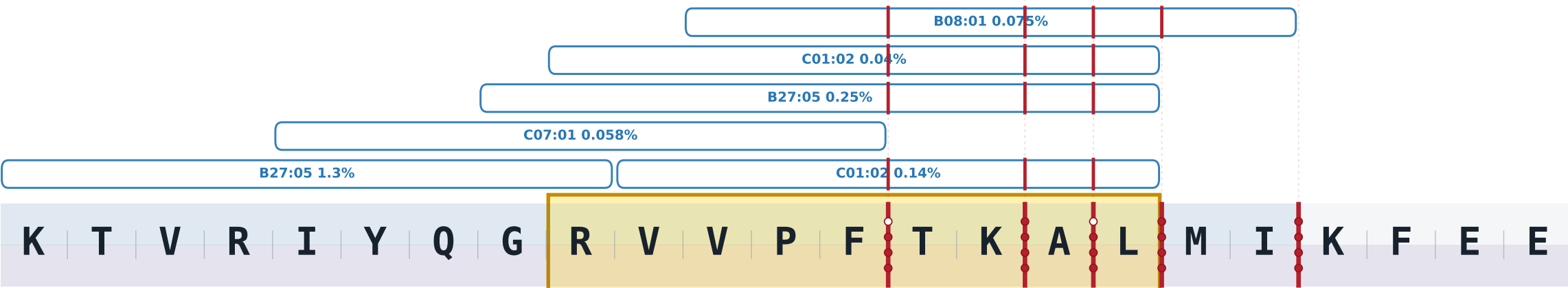
disclosed minimal epitope RVVPFTKAL at 9-17

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks



MHC-I: 6 shown / 35 eligible (top-10 cap)



MHC-II: 5 shown / 11 eligible (top-10 cap)

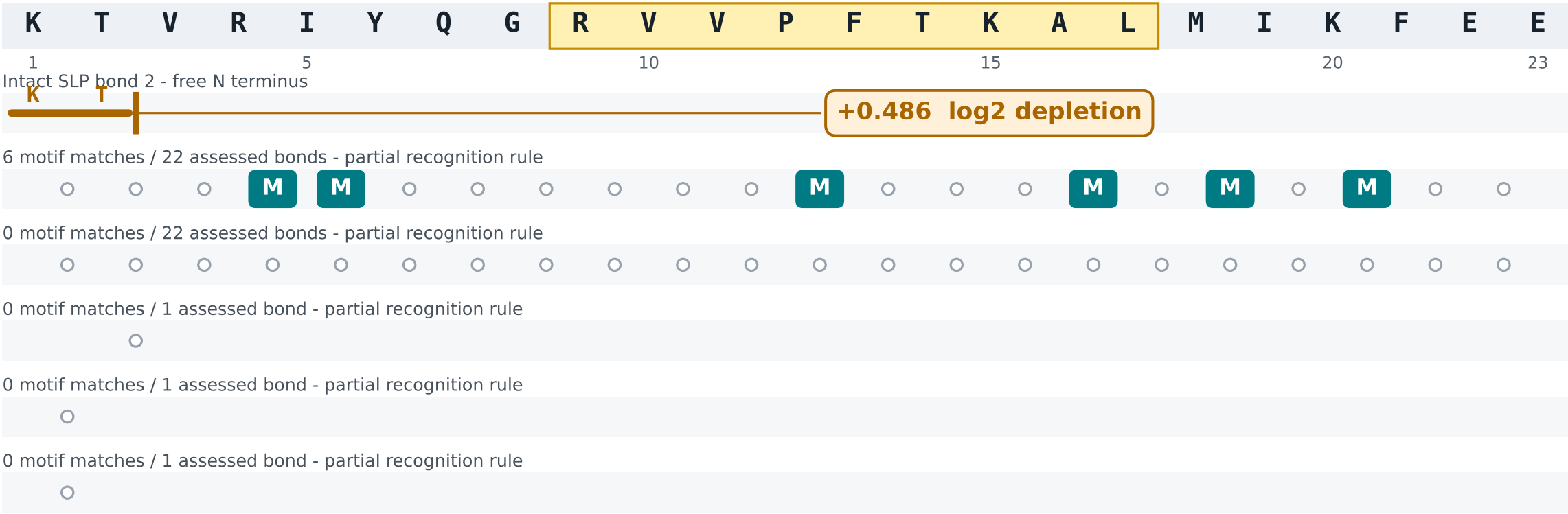


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.
DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

MAP2 p.Leu867fs | JLF V2, JLF V1 | 24 aa

disclosed minimal epitope RVVPFTKAL at 9-17

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

NetChop Cterm (ligand) native 0-1

Pepsickle human (in vivo) native 0-1

NetCleave I native 0-1

NetChop 20S (in vitro) native 0-1

0.5 display

0.5 display

0.5 display

0.5 display

MHC-I: 7 shown / 39 eligible (top-10 cap)

B08:01 0.13%

C01:02 0.04%

B27:05 0.25%

C07:01 0.058%

B27:05 1.3%

C01:02 0.14%

B08:01 0.096%

K T V R I Y Q G R V V P F T K A L M I K F E E I

DQ04:01/04:02 1.2%

DQ04:01/04:02 1.1%

DQ05:01/02:01 4.4%

DR08:01 3.5%

DP01:03/04:01 5%

NetCleave-II is not a named-cathepsin assay

NetCleave II native 0-1

0.5 display

EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure

K T V R I Y Q G R V V P F T K A L M I K F E E I

Intact SLP bond 2 - free N terminus

DPP4 / qPISA

N-terminal 2-residue removal

+0.486 log2 depletion

MME / neprilysin

Internal hydrophobic preference

FAP - internal

Gly-Pro|non-Pro recognition

FAP - N-terminal

Remove 2 residues: X-Pro|non-Pro

ANPEP / CD13

exposed N-terminus only

ENPEP

exposed N-terminus only

M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

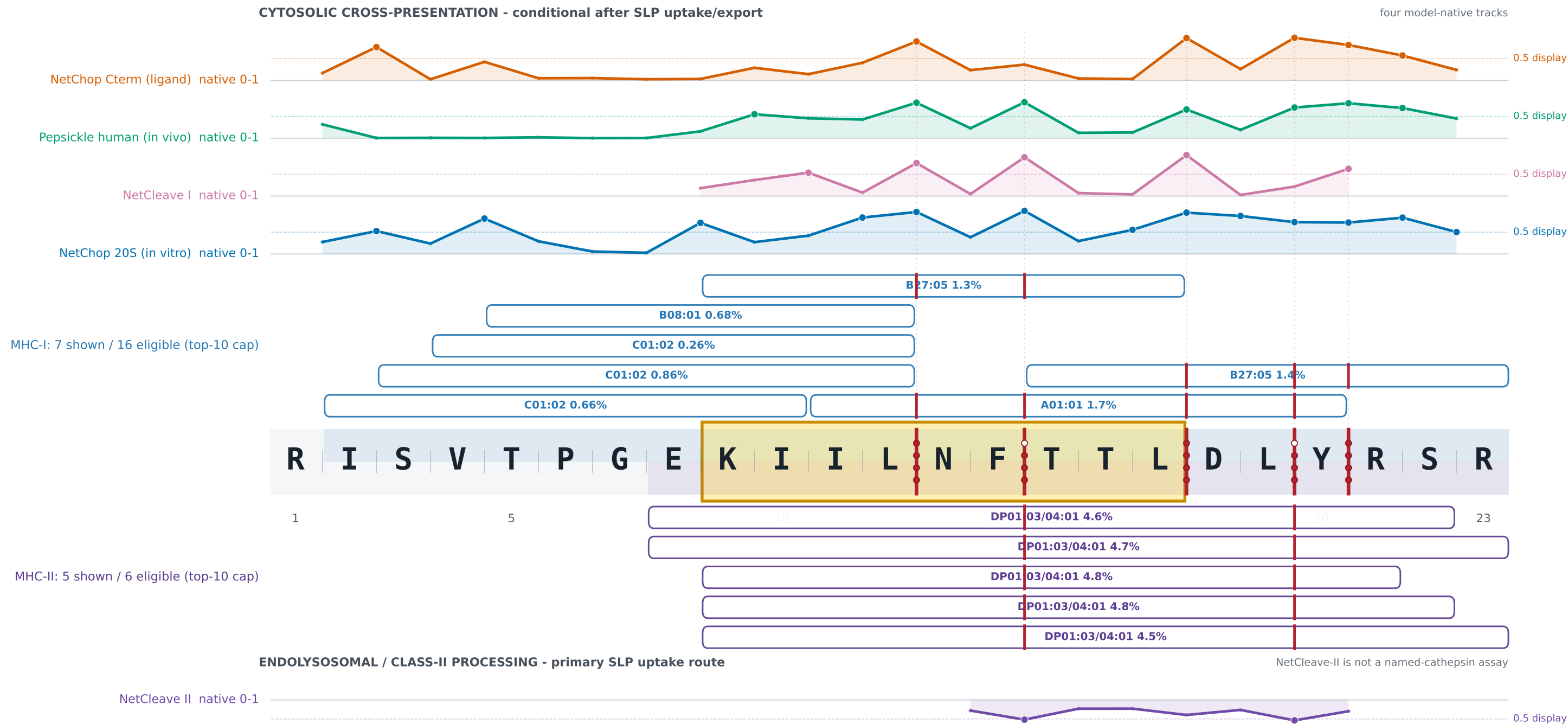
SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

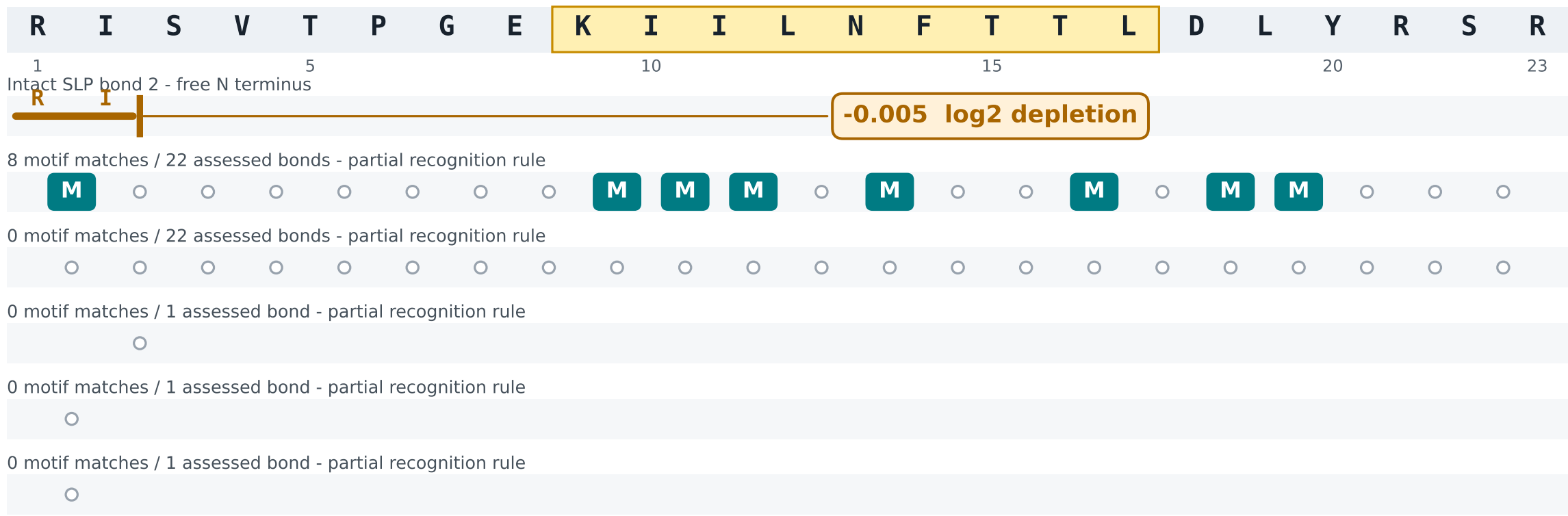
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

BMP1 p.Ser366Thr | JLF V3 | 23 aa

disclosed minimal epitope KIILNFTTL at 9-17



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

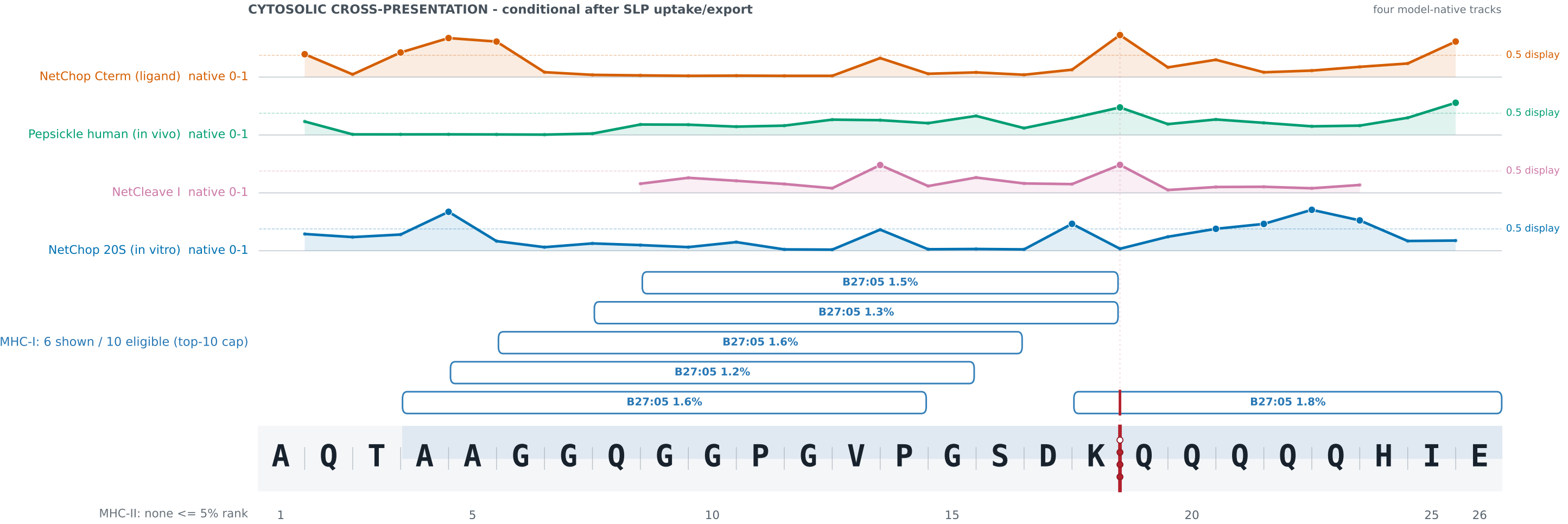
SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and ≥ 3 reached 0.5. Four filled/open beads show model concurrence, never probability.

MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

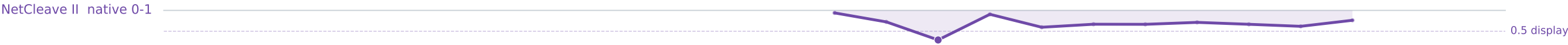
NR2F2 p.Gly65Val | JLF V2 | 26 aa

minimal epitope/window not disclosed for this SLP

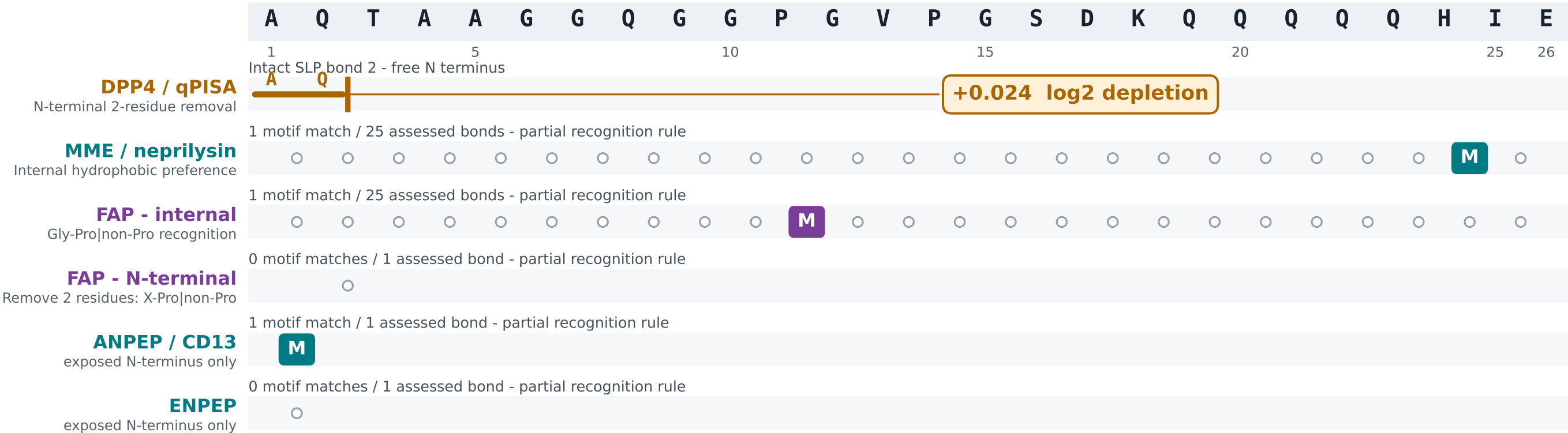


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

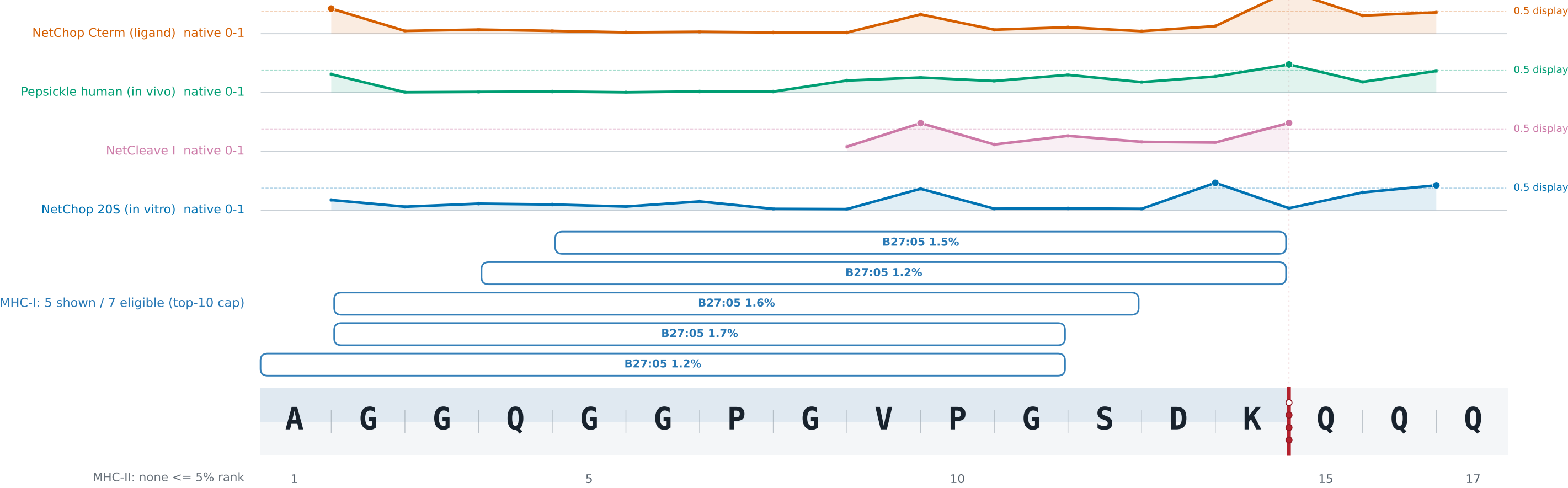
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

NR2F2 p.Gly65Val | CeGaT | 17 aa

minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

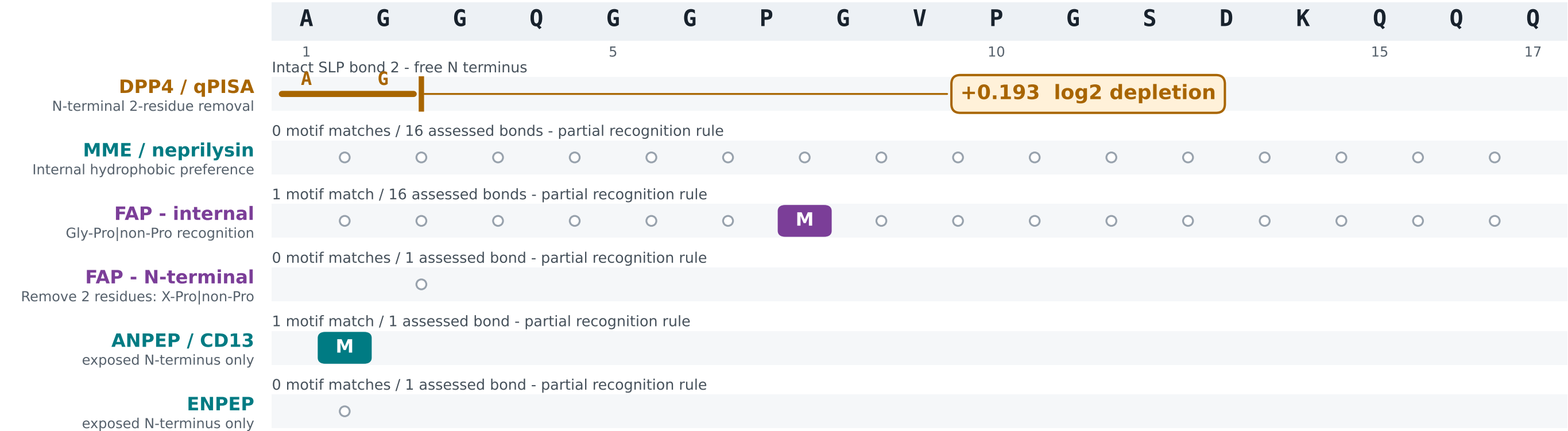


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay



EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

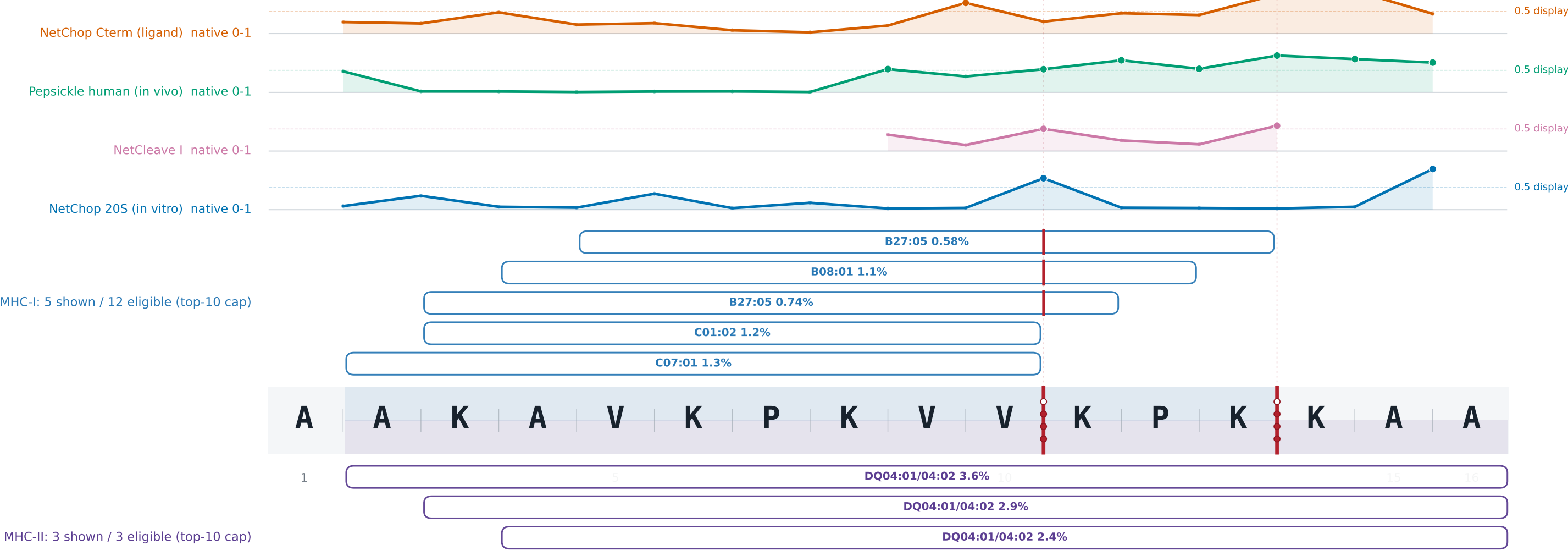
MHC Up to 10/class: intended overlap, then allele diversity, then native rank. Red ticks inside bars are pre-binding cut evidence; full results stay in CSV.

H1-2 p.Ala197_Lys201del | CeGaT | 16 aa

minimal epitope/window not disclosed for this SLP

CYTOSOLIC CROSS-PRESENTATION - conditional after SLP uptake/export

four model-native tracks

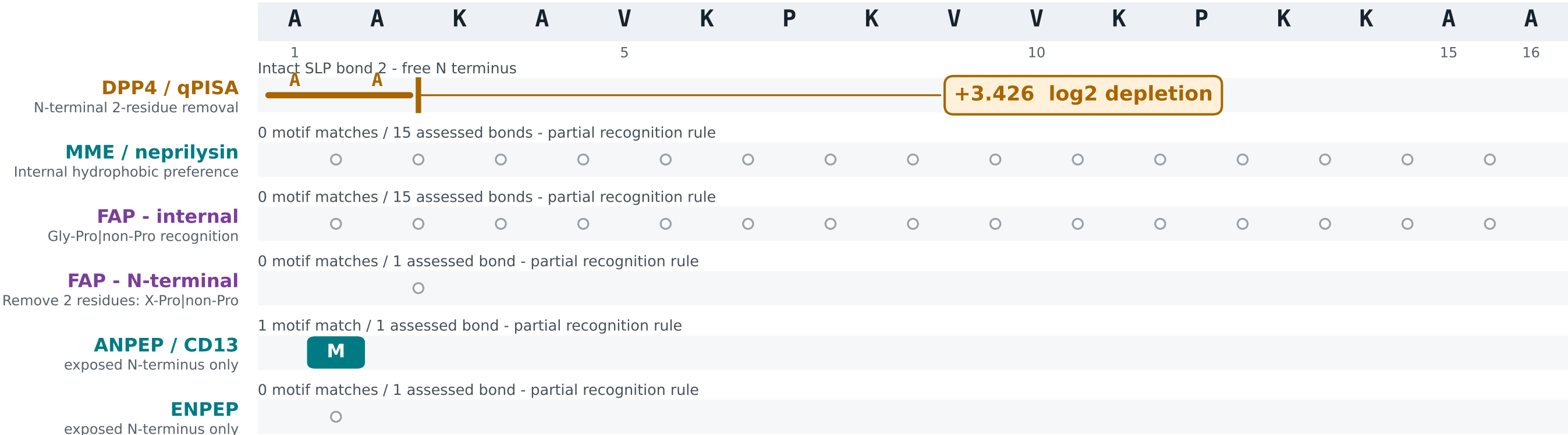


ENDOLYSOSOMAL / CLASS-II PROCESSING - primary SLP uptake route

NetCleave-II is not a named-cathepsin assay

NetCleave II native 0-1

EXTRACELLULAR ENZYME EVIDENCE - requires enzyme exposure



M = motif match | hollow circle = assessed, no motif match | NA = not assessed. Enzyme presence and exposure are not measured.

DPP4: predicted log2(buffer control / treated substrate), source assay 4 h. Motifs have no numeric score; neither view gives a cleavage rate.

SCORES Separate native 0-1 tracks; joined assessed bonds are not smoothed. Dashed = 0.5 display threshold; gaps = unassessed.

RED All four cytosolic tracks assessed the bond and >=3 reached 0.5. Four filled/open beads show model concurrence, never probability.

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